Chemistry

TARGET : JEE (Main + Advanced)

ENTHUSIAST + LEADER COURSE

Rankers Academy JEE

REVISION CLASS TEST # 01 to 14 (BOOKLET)

(INORGANIC)

(ENGLISH)

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SECTION-I (i) Single Correct (Maximum Marks:33)

Each question of this section have +3 marks.

1. Find the pair of species having the same shape but different hybridization of the central atom?

(A) $[SnCl_3]^{\ominus}$, XeO_3

(B) XeO_2F_2, SF_4

(C) XeF_2, $NO_2^{\oplus}$

(D) BrF_3 , $XeOF_2$

2. Select the **CORRECT** order of bond angle?

(A) $NH_3 < NH_2^{\ominus} < CH_4$

(B) $CO_2 = NO_2^{\oplus} < NO_2^{\ominus}$

(C) $SO_3^{-2} < SO_4^{-2} < SO_2$

(D) $BF_3 < BCl_3 < BBr_3$

3. Borazine shows a close similarity with benzene and therefore also named as inorganic benzene. Which of the following statement(s) is/are **INCORRECT** for benzene and borazine?

(A) Both are planar

(B) Both are aromatic

(C) Both are isoelectronic species

(D) Both have $2p\pi$ - $2p\pi$ type of coordinate bond.

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4. Identify the chemical specie(s), which is/are polar as well as planar.

(A) $SnCl_2$

(B) XeF_6

(C) BF_3

(D) XeF_2

5. With respect to valence bond theory, the correct statement is

(A) Bond energy order is $H_2 > F_2$

(B) Number of lone pair of electron order is $I_3^{\oplus} > I_3^{\ominus}$

(C) s % character order in lone pair of electron(s) is $OCl_2 > OF_2$

(D) Bond angle order is $SCl_2 > OCl_2$

6. Which of the following set of order is INCORRECT for their indicated properties?

(A) Bond length of O–F ; $O_2F_2 > OF_2$

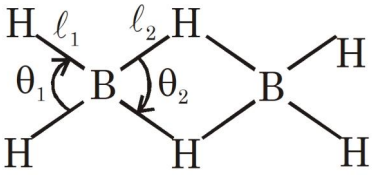
(B) Lewis acidic strength ; $BCl_3 > BF_3$

(C) Basic strength ; $N(CH_3)_3 < N(SiH_3)_3$

(D) $p_\pi - p_\pi$ back bond strength ; $BF_3 > BCl_3$

7. Observe the following conversion and structure of B_2H_6 (Diborane)

$2BH_{3(g)} \longrightarrow B_2H_{6(g)}$



The correct statement is :

(A) $\ell_1 > \ell_2$

(B) $\theta_1 > \theta_2$

(C) B_2H_6 consists 3c–2e bond and is a planar species

(D) The hybridisation state of Boron atom in B_2H_6 is not changed as compared to monomeric form of BH_3

8. Which of the following acid/base reaction is most difficult to proceed in forward direction ?

(A) $BCl_3 + NMe_3 \longrightarrow Cl_3B \leftarrow NMe_3$

(B) $BF_3 + NH_3 \longrightarrow F_3B \leftarrow NH_3$

(C) $NH_3 + H_2O \longrightarrow NH_4^+ + OH^-$

(D) $PH_3 + H_2O \longrightarrow PH_4^+ + OH^-$

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9. Which of the following species is having highest p% character in the orbital occupied by bond pair as compared to orbital occupied by lone pair ?

(A) NH_3 (B) PH_3

(C) AsH_3 (D) $\overline{CH_3}$

10. Which of the following species has bond angle $\geq 120°$, w.r.t. underlined atom?

(A) $\underline{O}(CH_3)_2$ (B) $\underline{N}(CH_3)_3$

(C) $\underline{N}(SiH_3)_3$ (D) $\underline{P}(SiH_3)_3$

11. The correct order of indicated bond length is :-

(A) d_{C-Cl} ; $CF_3-Cl > CH_3-Cl$

(B) d_{C-H} ; $CF_3-H < Cl_3C-H$

(C) d_{N-N} ; $H_2N-NH_2 < F_2N-NF_2$

(D) d_{O-O} ; $O_2F_2 > H_2O_2$

SECTION-I (ii) Multiple Correct : (Maximum Marks: 28)

Each question of this section have +4 marks.

12. Which of the following statement is/are correct?

(A) In NSF_3 both π-bonds are $p\pi$-$d\pi$ type.

(B) The ratio of σ-bonds to π-bond in SO_3 and SO_2 are identical.

(C) $[ICl_4]^{\ominus}$ & $[XeF_5]^{\ominus}$ are planar.

(D) $Cl_2O_{6(s)}$ exist as $ClO_2^{\oplus}$ & $ClO_4^{\ominus}$

13. Select the correct order of bond energy?

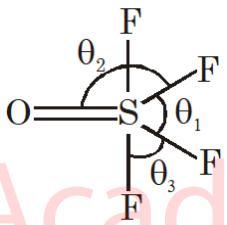
(A) H–F < H–Cl < H–Br

(B) $3p_\pi$-$3p_\pi$ < $3d_\pi$-$3d_\pi$

(C) $Cl_2 > Br_2 > F_2$

(D) $O_2 < O_3$(O–O)

14. Which of the following option is/are correct about



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(A) $\theta_2 > \theta_1 > \theta_3$

(B) $d_{x^2-y^2}$ orbital involved in hybridization

(C) maximum number of atoms in plane are 4.

(D) $S-F_{(axial)}$ bond length < $S-F_{(eq.)}$ bond length

15. Consider the following reaction and choose correct statements

$$3B_2H_6 + 6NH_3 \xrightarrow{\text{room temperature}} \text{3 mole ionic product (A)}$$

$$\text{3 mole ionic product (A)} \xrightarrow{\text{Heat}} \text{Product (B)} + \text{Flammable gas (C)}$$

(A) Product (B) contain aromatic character.

(B) Boiling point of product (C) is less than Helium.

(C) All Boron in product (A) are sp^3 hybridised.

(D) In product (B), $2p_\pi-3p_\pi$ type back bond present.

16. Which of the following is example of cyclic silicate?

(A) dioptase $Cu_6Si_6O_{18}.6H_2O$

(B) Crocidolite $Na_2Fe_5(OH)_2[(Si_4O_{11})]_2$

(C) Serpentine $Al_2(OH)_2Si_4O_{10}$

(D) Catapleite $Na_2ZrSi_3O_9.H_2O$

17. In which of the following odd electron species single electron is present in hybrid orbital.

(A) NO_2 (B) CF_3

(C) ClO_3 (D) ClO_2

18. Identify the pair(s) in which first chemical species has greater % s-character on lone pair of central atom than second one.

(A) (NH_3, NF_3)

(B) (PF_3, PH_3)

(C) $(SnCl_2, SnCl_3^-)$

(D) (SbH_3, PH_3)

SECTION-III Integer : (Maximum Marks: 9)

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Each question of this section have +3 marks.

1. Identify the pair in which the specified bond length of first is greater than second.

PCl_3F_2 , PF_3Cl_2 : B L_{P-Clcq}

SO_2Cl_2 , SO_2F_2 : B $L_{S=O}$

BF_3 , BCl_3 : B L_{B-X} X = F/Cl

NO_3^- , NO_2^- : B L_{N-O}

O_3 , O_2 : B L_{O-O}

CO , CO_2 : B L_{C-O}

2. Find the total number of chemical specie(s) in which effective $p\pi - d\pi$ type of back bonding is observed.

$N(SiH_3)_3$, $O(SiH_3)_2$, $:CCl_2$, $P(CH_3)_3$, $N(GeH_3)_3$, $B(OMe)_3$, $B(OEt)_3$, SiF_4 , $P(SiH_3)_3$

3. Write sum of basicity of oxy acids formed by hydrolysis of

$PCl_3 \xrightarrow{\text{Hydrolysis}} \underset{\text{(I)}}{\text{oxy acid}} + \text{hydra acid}$

$SF_4 \xrightarrow{\text{Hydrolysis}} \underset{\text{(II)}}{\text{oxy acid}} + \text{hydra acid}$

$BrF_5 \xrightarrow{\text{Hydrolysis}} \underset{\text{(III)}}{\text{oxy acid}} + \text{hydra acid}$

SECTION-IV Match the Column : (Maximum Marks: 8)

Each question of this section have +8 marks
(Partial marks of each correct is +2)

1.

	Column-I (Treatment of reactant with excess water at room temperature)		Column-II (Reaction Characteristics)
(A)	$XeF_4 \xrightarrow{H_2O}$	(P)	Disproportionation reaction
(B)	$XeF_6 \xrightarrow{H_2O}$	(Q)	One of the product is oxygen gas
(C)	$SF_6 \xrightarrow{H_2O}$	(R)	Complete hydrolysis
(D)	$N_2O_4 \xrightarrow{H_2O}$	(S)	Inert towards hydrolysis
		(T)	One of the product is monobasic oxyacid

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SECTION-I (i) Single Correct (Maximum Marks:33)

Each question of this section have +3 marks.

1. Which of the following exhibit sp mixing ?

(A) O_2^{2+} (B) O_2

(C) N_2^{2-} (D) All of the above

2. Which of the following is paramagnetic & B.O. is not in fractional value

(A) NO (B) N_2^{2+}

(C) N_2^{2-} (D) All of the above

3. Which of the following species contain fractional bond order.

(A) H_2 (B) NO

(C) F_2 (D) N_2

4. Which of the following have intra molecular H-bonding.

(A) Salicylic acid (B) o-nitrophenol

(C) Salicylaldehyde (D) All of these

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5. The correct order of boiling point of the following pair of species is :

(A) $CCl_4 < SiCl_4$ (B) $NF_3 < NMe_3$

(C) $ICl < Br_2$ (D) $D_2O < H_2O$

6. Consider the following four xenon compounds: XeF_4, XeF_2, XeO_3, XeO_2F_2.
The pair of xenon compounds expected to have non-zero dipole moment is

(A) XeF_4 and XeO_3 (B) XeO_3 and XeF_2

(C) XeO_3 and XeO_2F_2 (D) XeO_2F_2 and XeF_4

7. The potential energy of intermolecular interaction in solid ICl is proportional to

(A) $\frac{1}{r^4}$ (B) $\frac{1}{r^6}$

(C) $\frac{1}{r^3}$ (D) $\frac{1}{r}$

8. Which intermolecular force is most responsible in allowing Xenon gas to liquify?

(A) Ionic (B) London dispersion force

(C) Dipole-dipole (D) Ion-dipole

9. Which one of the following alkaline earth metal sulphate has largest cation :-

(A) $BaSO_4$ (B) $SrSO_4$

(C) $CaSO_4$ (D) $BeSO_4$

10. The correct order of C–O bond length in $CO_3^{-2}, CO_2, HCO_2^-, CO$:-

(A) $CO > CO_3^{-2} > HCO_2^- > CO_2$

(B) $CO_3^{-2} > HCO_2^- > CO_2 > CO$

(C) $HCO_2^- > CO_2 > CO_3^{-2} > CO$

(D) $CO_3^{2-} > CO_2 > CO > HCO_2^-$

11. Which of the following set contain peroxy linkage :-

(A) $HNO_4, H_2S_2O_8$ (B) $H_2S_2O_5, H_4P_2O_7$

(C) $H_2S_2O_7, H_4P_2O_8$ (D) $H_4P_2O_5, H_4P_2O_6$

SECTION-I (ii) Multiple Correct : (Maximum Marks: 12)

Each question of this section have +4 marks.

12. Identify the incorrect statement(s) for the hydrolysis of P_4O_{10} when treated with excesss of water.

(A) Tetrametaphosphorous acid is formed as one of product during process

(B) Final product is tribasic oxyacid

(C) One of the intermediate product has four P–O–P linkage

(D) Maximum basicity of intermediate product is 4 (out of all intermediate products formed in reaction)

13. Which of the following compounds show intermolecular hydrogen bonding?

(A) Ortho chlorophenol

(B) Chloral

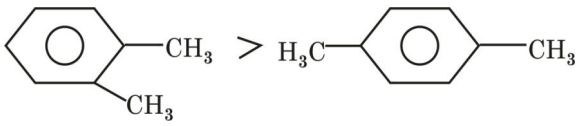
(C) Meta hydroxybenzaldehyde

(D) Para hydroxybenzaldehyde

14. Select the correct order of boiling point?

(A) $BMe_3 > BF_3$

(B) iso-pentane > neo-pentane

(C) 

(D) $D_2O > H_2O$

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SECTION-I (iii) Paragraph (SOT) : (Maximum Marks: 12)

Each question of this section have +3 marks.

Paragraph for Questions 15 and 16

The cation polarises the anion, pulling the electronic charge toward itself and thereby increasing the electronic charge between the two. This is precisely what happens in a covalent bond, i.e., buildup of electron charge density between the nuclei. The polarising power of the cation, the polarisability of the anion and the extent of distortion (polarisation) of anion are the factors, which determine the per cent covalent character of the ionic bond.

Polarisation of anion is used to compare the covalent and ionic character of molecule, compare the nature of oxide, compare the electrical conductivity of ionic comopounds, tendency of the formation of complex compounds, compare the thermal stability of metal salts, compare the intensity of colour of compounds, compare the solubility of heavier metal halide in water.

15. Identify the incorrect order

(A) $AgF < AgCl < AgBr$ (solubility in water)

(B) $NaF < MgF_2 < AlF_3$ (melting point)

(C) $ZnS < CdS < HgS$ (intensity of colour)

(D) $BeO < MgO < CaO$ (basic character)

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16. Identify the pair in which first cation having greater complex formation tendency.

(A) (Ba^{+2}, Be^{+2}) (B) (Fe^{+2}, Fe^{+3})

(C) (Be^{+2}, Li^{+}) (D) (Ca^{+2}, Zn^{+2})

Paragraph for Questions 17 and 18

In the heavier elements of p-block elements, however as we expect form the "inert pair effect, the lower oxidation state becomes more common on descending the group."

Covalent nature in ionic compound is explained on the basis of polarisation effect.

17. Which of the following compound is most covalent nature?

(A) $SnCl_2$ (B) $CaCl_2$

(C) $SnCl_4$ (D) $InCl_3$

18. The correct order of stability is :

(A) $Sn^{2+} > Sn^{4+}$ (B) $Pb^{2+} > Pb^{+4}$

(C) $Tl^{+3} > Tl^{\oplus}$ (D) $Bi^{3+} < Bi^{5+}$

SECTION-II Integer : (Maximum Marks: 6)

Each question of this section have +3 marks.

1. Find the sum of bond order in the given species having fractional bond order.
NO, O_2^-, N_2, $H_2^{\oplus}$, CO, C_2^{-2}, O_2, CO_3^{-2}

2. Find the number of chemical species having greater boiling point than He
H_2 , Ne , Ar , Kr , SiO_2, NaCl, T_2, D_2

SECTION-IV Match the Column : (Maximum Marks: 8)

Each question of this section have +8 marks
(Partial marks of each correct is +2)

1.

	Column-I Oxy acid		Column-II
(A)	Pyrosulphurous acid	(P)	Basicity = 2
(B)	Peroxydisulphuric acid	(Q)	S–S linkage is present
(C)	Dithionic acid	(R)	Difference in oxidation state of sulphur atoms is zero
(D)	Pyrosulphuric acid	(S)	S–O–S linkage is present
		(T)	Atleast one sulphur atom have +3 oxidation state

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SECTION-I (i) Single Correct (Maximum Marks:33)

Each question of this section have +3 marks.

1. The given complexes can exhibit which type of isomerism respectively
$[Cr(H_2O)_3(NH_3)_3]^{+3}$, $[Cr(en)_3]^{+3}$ and $[Cr(Gly)_3]^0$

(A) (Optical isomerism & Geometrical isomerism), (Optical isomerism) and (Optical isomerism & Geometrical isomerism)

(B) (Geometrical isomerism), (Optical isomerism & Geometrical isomerism) and (Geometrical isomerism)

(C) (Geometrical isomerism), (Optical isomerism) and (Geometrical isomerism & Optical isomerism)

(D) (Optical isomerism & Geometrical isomerism), (Optical isomerism & Geometrical isomerism) and (Optical isomerism)

2. The oxidation state of Cr in $[Cr(H_2O)_6]Cl_3$, $[Cr(C_6H_6)_2]$ and $K_2[Cr(CN)_2O_2(O_2)(NH_3)]$ respectively are.

(A) +3, +2, +4 (B) +3, 0, +6

(C) +3, 0, +4 (D) +3, +4, +6

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3. Choose incorrect option

(A) E.A.N. for $\underline{Fe_2}(CO)_9$ is 36 (B) E.A.N. for $\underline{Mn_2}(CO)_{10}$ is 36

(C) E.A.N. for $\underline{[Ag(CN)_2]^-}$ is 54 (D) $Fe_2(CO)_9$ has one Fe–Fe bonds

4. For $Fe_4[Fe(CN)_6]_3$ which option is not correct

(A) IUPAC name is Iron(III) hexacyanidoferrate(II)

(B) It has total 12 π bonds

(C) It has coordination number is 6

(D) Ionisation sphere has Fe^{3+} & coordination sphere has Fe^{2+} ions are present.

5. The number of geometrical isomers possible for complex $[M(AB)_2C_2]^{\pm n}$ will be

(A) 4 (B) 5

(C) 3 (D) 6

6. In [M(EDTA)Cl] type octahedral complex has how many O-M-N bonds are present
(where EDTA = ethylenediaminetriacetate ion and M = metal) :-

(A) 6 (B) 8

(C) 4 (D) None

7. Compare C–O & M–C bond length in the following complexes.
a) $[M(CO)_x]^{2+}$ b) $[M(CO)_x]^{+}$ c) $[M(CO)_x]^{0}$ d) $[M(CO)_x]^{-}$ e) $[M(CO)_x]^{2-}$

(A) for C–O a < b < c < d < e ; for M–C a > b > c > d > e

(B) for C–O a > b > c > d > e ; for M–C a > b > c > d > e

(C) for C–O a > b > c < d < e ; for M–C a < b < c > d > e

(D) for C–O a > b > c > d > e ; for M–C a < b < c < d < e

8. Complex $CoClBr \cdot 5NH_3$ has two isomers I and II. Isomer I and II have Cl^- and Br^- in their co-ordination sphere respectively. Choose incorrect statement :-

(A) Both isomers are ionisation isomers of each other

(B) Both isomers do not show geometrical isomerism

(C) Primary valency of both isomers is satisfied by same ions.

(D) Isomer I produce white ppt. on reaction with $AgNO_3$(aq.)

9. In the isoelectronic series of metal carbonyl, the M–C π-bond strength order is

(A) $[Mn(CO)_6]^+ < [Cr(CO)_6] < [V(CO)_6]^-$

(B) $[V(CO)_6]^- < [Cr(CO)_6] < [Mn(CO)_6]^+$

(C) $[V(CO)_6]^- < [Mn(CO)_6]^+ < [Cr(CO)_6]$

(D) $[Cr(CO)_6] < [Mn(CO)_6]^+ < [V(CO)_6]^-$

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10. Which of the following complex show geometrical and optical isomerism both :-

(A) $[Co(ox)_3]^{3-}$ (B) $[Co(en)_2Cl_2]^+$

(C) $[Mn(NH_3)_4Cl_2]^{+1}$ (D) $[CoCl_2Br_2]^{2-}$

11. Complex in which number of trans isomers with respect to 'aa' are maximum :-

(A) Ma_2bcde (B) $Ma_2b_2c_2$

(C) Ma_2b_2cd (D) Ma_2b_4

SECTION-I (ii) Multiple Correct : (Maximum Marks: 16)

Each question of this section have +4 marks.

12. Which of the following ligands are symmetrical based on denticity or structure?

(A) en (B) bn

(C) gly^- (D) $C_2O_4^{-2}$

13. Which of the following complex obey sidwick EAN rule.

(A) $[Fe(\eta^5\text{-}C_5H_5)_2]$ (B) $[Ni(DMG)_2]$

(C) $[Mn_2(CO)_{10}]$ (D) $[Co_2(CO)_8]$

14. Which of the following complexes shows geometrical isomerism as well as optical isomerism ?

(A) $[Ni(en)_3]^{2\oplus}$ (B) $[PtCl_2(PPh_3)(NH_3)]$

(C) $[Fe(en)_2(CN)_2]^{\oplus}$ (D) $[Co(ox)Cl_2F_2]^{3\ominus}$

15. Which of the following statement is/are correct?

(A) [Mabcd] type of square planar complex shows three geometrical isomers – two cis and one trans.

(B) [M(EDTA)] type of octahedral complex have two stereoisomers.

(C) $[Ma_3b_3]$ type of octahedral complex have three stereoisomers.

(D) $[M(bn)_2]$ type of square planar complex shows geometrical isomerism and optical isomerism.

SECTION-I (iii) Paragraph (MCQ) : (Maximum Marks: 8)

Each question of this section have +4 marks.

Paragraph for Questions 16 and 17

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Complex compounds that have same molecular formula but have different structural or space arrangements of ligand around central metal atom/ion are called isomers, these are two types namely structural and stereoisomers.

16. Which type of isomerism is/are shown by complex ion $[CrCl_2(NO_2)_2(NH_3)_2]^-$

(A) Geometrical isomerism

(B) Optical isomerism

(C) Linkage isomerism

(D) Coordinate isomerism

17. Which of the following complex compound/s does not exhibit geometrical isomerism.

(A) $[CoCl_2Br_2]^{2-}$

(B) $[Be(gly)_2]$

(C) $[Pt(NH_3)_2NO_2Cl]$

(D) $[Co(en)_3]^{+3}$

SECTION-III Integer : (Maximum Marks: 6)

Each question of this section have +3 marks.

1. Find the total number of five membered ring in $[Co(EDTA)]^{-1}$ complex ion?

2. Find the number of ligands which are neutral & bidentate ligands among the following –
(a) en (b) bn (c) tn (d) pn (e) dimethylglyoximato (f) ox
(g)acac (acetylacetanato) (h) ethylenediaminetetraacetato

SECTION-IV Match the Column : (Maximum Marks: 8)

Each question of this section have +8 marks
(Partial marks of each correct is +2)

1. Match the column :

	Column-I Complex compound		Column-II Characteristics
(A)	$[Co(Cl)(CN)(NO_2)(NH_3)_3]$	(P)	Shows linkage isomerism
(B)	$Mn_2(CO)_{10}$	(Q)	No geometrical isomers
(C)	$K[Co(CO)_4]$	(R)	EAN value of central metal is 36
(D)	$[Cr(C_2O_4)_3]^{-3}$	(S)	Oxidation number of metal ≤ 0
		(T)	Metal–Metal bond

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SECTION-I (i) Single Correct (Maximum Marks:21)

Each question of this section have +3 marks.

1. Which of the following pair of complex/complex-ion is correct against to their indicated properties?

(A) $[Ni(CN)_4]^{2\ominus} > [NiF_6]^{4\ominus}$; C.F.S.E.

(B) $[Ni(CO)_4] < [Fe(CO)_5]$; Valence shell electrons on central metal ion

(C) $[Fe(CN)_6]^{3\ominus} > [FeF_6]^{3\ominus}$; Number of unpaired electrons of central metal ion

(D) $[Co(ox)_3]^{3\ominus} > [Co(CN)_6]^{3\ominus}$; Stability

2. Complex which does not follow sidwick EAN rule :-

(A) Ferrocene

(B) Brown ring complex

(C) $K_2[PtCl_6]$

(D) $[Fe(CO)_2(NO)_2]$

3. Which wavelength of visible light can be absorbed by $[Cu(H_2O)_4]SO_4.H_2O$ and $[Fe(H_2O)_6]SO_4.H_2O$ complex respectively.

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(A) Yellow and Voilet

(B) Orange and Red

(C) Blue and Green

(D) Green and Orange

4. CFSE value and spin only magnetic moment for $[RhF_6]^{3-}$ are _____and _____.

(A) $-2.4\ \Delta_0 + 2P$, 0 B.M.

(B) $-2.4\ \Delta_0 + 3P$, $\sqrt{24}$ B.M.

(C) $-0.4\ \Delta_0 + 2P$, $\sqrt{24}$ B.M.

(D) $-0.4\ \Delta_0 + 3P$, $\sqrt{24}$ B.M.

5. Which of the following complex is diamagnetic as well as inner orbital complex?

(A) $[Co(OH_2)_6]SO_4$

(B) $K_4[Fe(CN)_5(O_2)]$

(C) $[Mn(NCS)_6]^{-4}$

(D) $[Co(NH_3)_6]Cl_3$

6. Which of the following statements is/are correct?

(A) $[CoF_6]^{3\ominus}$ ion is low spin complex due to pairing energy > Δ_0

(B) In Zeise's salt, back donation weakens the double bond of alkene.

(C) Among Ar, NaCl and HCl, the NaCl shows weakest inter ionic interaction.

(D) $O_2^{2\ominus}$ and $NO^{\oplus}$ have same bond order.

7. $\overset{(X)}{\text{Salt}} + Na_2[Fe(CN)_5\overset{\oplus}{N}O] \xrightarrow{\text{basic solution}} \text{Purple Colour}$

Which of the following statement is correct for purple colour complex?

(A) Denticity of new ligand formed is 3

(B) It's magnetic nature is paramagnetic

(C) It is low spin complex

(D) Hybridisation of Fe is sp^3d^2

SECTION-I (ii) Multiple Correct : (Maximum Marks: 12)

Each question of this section have +4 marks.

8. Which of the following statement is correct against to the indicated properties?

(A) $[Pt(NH_3)_4]Cl_2 > K_2[PtCl_4]$; Crystal field splitting energy

(B) $[Co(NH_3)_3ClBr(NO_3)] > [Co(NH_3)_3Cl_3]$; Number of geometrical isomers

(C) $K_4[Fe(CN)_6] > K_3[Fe(CN)_6]$; Valence shell's electrons of central metal ion

(D) $[Fe(CO)_2(NO)_2] > [Fe(edta)]^{\ominus}$; Number of Fe–N linkages.

9. Which of following species is/are paramagnetic?

(A) $NO^{\ominus}$

(B) $[NiCl_4]^{2\ominus}$

(C) $[CuCl_4]^{3\ominus}$

(D) $O_2[AsF_6]$

10. Which of the following order is/are correct for stability?

(A) $[CoF_6]^{-3} < [Co(C_2O_4)_3]^{-3} < [Co(CN)_6]^{-3}$

(B) $[Os(H_2O)_6]^{+2} < [Ru(H_2O)_6]^{+2} < [Fe(H_2O)_6]^{+2}$

(C) $[Co(H_2O)_6]^{+3} < [Co(CN)_6]^{-3} < [Ir(CN)_6]^{-3}$

(D) $[NiCl_4]^{-2} < [Ni(NH_3)_6]^{+2} < [Ni(CN)_4]^{-2}$

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SECTION-I (iii) Paragraph (SOT) : (Maximum Marks: 6)

Each question of this section have +3 marks.

Paragraph for Questions 11 and 12

Addition compound which retain their identity in solution are termed as co-ordination compound. Bonding in co-ordination compound is explained by VBT & CFT. Co-ordination compound also show both structure and stereoisomerism.

11. Which of the following complex having maximum number of optically active isomer.

(A) $[Ma_2bcde]^{n\pm}$

(B) $[M(AB)c_2d_2]^{n\pm}$

(C) $[Ma_2b_2c_2]^{n\pm}$

(D) $[M(AB)_2(CC)]^{n\pm}$

12. Identify the **INCORRECT** matching in the following.

	Complex	Isomerism	Complex type
(1)	$[Co(H_2O)_6]Cl_2$	Show structural isomersim	low spin
(2)	$K_3[Co(ox)_3]$	Show optical isomerism	low spin
(3)	$[Pt(NH_3)_2Cl_2Br_2]$	Form 6 stereo isomers	low spin
(4)	$[Cu(NO_2)_4]^{2-}$	Show structural isomerism	square planar

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SECTION-I (iv) Matching List : (Maximum Marks: 3)

Each question of this section have +3 marks

13. Match list I with list II and select the correct answer?

	List-I (Complex compound)		List-II (Characteristics)
(P)	$[CoF_3(H_2O)_3]$	(A)	Hybridisation of central metal ion is sp^3d^2
(Q)	$[Co(NO_2)_6]^{-4}$	(B)	Paramagnetic
(R)	$[Cr(CN)_3(H_2O)_3]$	(C)	Inner orbital complex
(S)	$[Zn(NO_2)_4]^{-2}$	(D)	$\mu_m = 0$ B.M.

The correct option is

(A) P $\longrightarrow$ 1, 3 ; Q $\longrightarrow$ 3, 4 ; R $\longrightarrow$ 2, 3 ; S $\longrightarrow$ 2, 4

(B) P $\longrightarrow$ 1, 2 ; Q $\longrightarrow$ 2 ; R $\longrightarrow$ 2, 4; S $\longrightarrow$ 4

(C) P $\longrightarrow$ 1, 2 ; Q $\longrightarrow$ 2, 3 ; R $\longrightarrow$ 2, 3 ; S $\longrightarrow$ 4

(D) P $\longrightarrow$ 1, 4 ; Q $\longrightarrow$ 2, 3 ; R $\longrightarrow$ 2, 3 ; S $\longrightarrow$ 4

SECTION-II Integer : (Maximum Marks: 15)

Each question of this section have +3 marks.

1. Find the number of species which are diamagnetic and square planar geometry.
$[IrCl(PPh_3)_3]$, $[RhCl(PPh_3)_3]$, trans-platin, $[Co(edta)]^{\ominus}$, $[NiCl_4]^{2\ominus}$, $[Cu(CN)_4]^{3\ominus}$, $[Ni(CO)_4]$, $[AuCl_4]^{\ominus}$ [Z of Ir = 77, Rh = 45, Pt = 78, Co = 27, Ni = 28, Cu = 29, Au = 79]

2. Find the number of the paramagnetic and inner orbital complex in the following :-
$[Ni(CO)_4]$, $[Ni(CN)_4]^{2-}$, $[Co(H_2O)_6]^{3+}$, $[Cu(NH_3)_4]^{2+}$, $[Cr(NH_3)_6]^{3+}$, $[Fe(CN)_6]^{3-}$

3. Total number of electrons in t_{2g} orbital of Sodium nitropruside i.e. $Na_2[Fe(CN)_5NO]$ = X and the total number of electrons in t_{2g} orbital of brown ring complex i.e. $[Fe(H_2O)_5NO]^{+2}$ = Y
then find value of $\frac{X + Y}{2}$

4. $$[Fe(H_2O)_6]^{3+}_{(aq.)} + \underset{(excess)}{SCN^{\ominus}_{(aq.)}} \longrightarrow \underset{\text{(Blood red colour solution)}}{[Fe(H_2O)_3(SCN)_3]}$$
$$\downarrow + F^{\ominus}_{(aq.)} \text{ (excess)}$$
$$\text{Colourless complex [Y]}$$

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The spin magnetic moment of hexa fluoro complex (Y) is :

5. "Crystal field splitting energy" (CFSE = "Δ_O") for $[Ti(H_2O)_6]^{3+}$ is 242 KJ/mol.
"Crystal field stabilisation energy" in KJ/mol is :

SECTION-I (i) Single Correct (Maximum Marks:42)

Each question of this section have +3 marks.

1. In which of the following reactions, Na_2O is not formed ?

(A) $Na_2O_2 + Na \longrightarrow$ (B) $NaNO_2 + Na \longrightarrow$

(C) $Na_2O_2 + SO_2 \longrightarrow$ (D) $NaNO_3 + Na \longrightarrow$

2. By product of 'Solvay process' is:

(A) $CaCl_2$ (B) NH_4Cl

(C) CO_2 (D) NH_3

3. Which of the reaction is **INCORRECTLY** written with NH_3 in different condition (Liq. NH_3 ; gas NH_3 ; NH_3 (aq.) solution)

(A) $Na + NH_3(\ell) \longrightarrow Na^+ (NH_3)_x + e^-(NH_3)_y$

(B) $Pb^{2+} + 4NH_3aq \longrightarrow [Pb(NH_3)_4]^{2+}$

(C) $2Na + 2NH_3(g) \longrightarrow 2NaNH_2 + H_2(gas) \uparrow$

(D) $AgNO_3 + \underset{\text{(excess)}}{2NH_4OH(\text{aq. solution of } NH_3)} \rightarrow [Ag(NH_3)_2]^+ + NO_3^- + 2H_2O$

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4. Which of the following ion is unstable in aqueous solution?

(A) Co^{+2} (B) Fe^{+2}

(C) Mn^{+2} (D) Cu^{+1}

5. Slaked lime is obtained when water is added to :

(A) $CaSO_4 \cdot \frac{1}{2}H_2O$ (B) $CaCl_2$

(C) CaO (D) $CaCO_3$

6. Choose correct statement

(A) In $(NH_4)_2Cr_2O_7$ geometry around Nitrogen & each Chromium is not same

(B) On heating $(NH_4)_2\ Cr_2O_7$ give N_2 as a product and it is endothermic reaction.

(C) All Nitrate salt on decomposition gives atleast one gaseous product

(D) Metal nitrates are generally insoluble in water.

7. Metal hydroxide $\xrightarrow[\text{passed}]{CO_2 \text{ is}}$ Metal carbonate $\xrightarrow[\text{is passed}]{\text{excess } CO_2}$ metal bicarbonate

Which of the following metal hydroxide produce soluble carbonate as well as soluble bicarbonate in the above reaction sequence.

(A) $Ca(OH)_2$ (B) NH_4OH

(C) $Ba(OH)_2$ (D) KOH

8. $BeCl_2 + LiAlH_4 \longrightarrow X + LiCl + AlCl_3$

Total number of $3c-4e^{\ominus}$ bond present in monomer form of compound (X)

(A) 1 (B) 2

(C) 3 (D) 0

9. $KO_2 + H_2O \longrightarrow KOH + O_2 + \text{product (X)}$

Choose the correct about product (X)

(A) It has open book like structure

(B) It stabilized by urea

(C) It decomposes in presence of light

(D) All of these

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10. Which of the following metal does not react with nitrogen and cannot form nitride

(A) Al (B) Mg

(C) Cs (D) Li

11. Consider the reaction

$K_2[Ni(CN)_4] + NH_3 + 2K(\text{potassium}) \longrightarrow$ complex (A)

complex (A) gives five ions in aqueous solution then choose correct statements.

(A) complex is sp^3 hybridised & diamagnetic

(B) complex is sp^3 hybridised and paramagnetic

(C) complex is dsp^2 hybridised and diamagnetic

(D) In complex $\overset{\ominus}{C}N$ ligands are replaced by NH_3 ligands

12. The electronic configuration of central atom of complex $[MnO_4]^{2-}$ is

(A) $t_2^1\, e^0$ (B) $t_2^0\, e^0$

(C) $e^0\, t_2^0$ (D) $e^1\, t_2^0$

13. On heating 'Trona ($Na_2CO_3.NaHCO_3.2H_2O$)' with the help of Bunsen Burner, we get _____ as solid residue

(A) Na_2CO_3

(B) Na_2O

(C) $Na_2CO_3.H_2O$

(D) $NaHCO_3$

14. Gypsum is added into the Portland Cement to

(A) Stop setting process

(B) Slow down the setting process

(C) Fasten the setting process

(D) Adjust the viscosity of the slurry

SECTION-I (ii) Multiple Correct : (Maximum Marks: 20)

Each question of this section have +4 marks.

15. Which of the following statement(s) is/are correct.

(A) Among alkaline earth metal oxalate, MgC_2O_4 is least soluble in water

(B) Among alkali metal carbonate Li_2CO_3 is least thermally stable.

(C) KO_3 is paramagnetic and orange red coloured

(D) KO_2 is used as purifier in ill ventilated room

16. Which of the following reactions give correct product?

(A) $Mn^{2+} + (NH_4)_2S_2O_8 + H_2O \longrightarrow MnO_4^{2-} + SO_4^{2-} + NH_4^+ + H^+$

(B) $K^+ + NO_2^- + Co^{2+} \longrightarrow K_4[Co(NO_2)_6] + NO + H_2O$

(C) $(NH_4)_2Cr_2O_7 \xrightarrow{\Delta} N_2 + Cr_2O_3 + 4H_2O$

(D) $Cu + 2H_2SO_4$ (conc.) $\longrightarrow CuSO_4 + SO_2 + 2H_2O$

17. Choose the incorrect order of the indicated property given in the options.

(A) He < Ne < Kr < Ar : Clathrate compound formation tendency order

(B) $NaO_2 < KO_2 < RbO_2 < CsO_2$: Thermal stability order

(C) $BeF_2 < BaF_2$: Solubility order in water

(D) LiF > LiCl > LiI : Melting point order

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18. Which of the properties are incorrect regarding $NaHCO_3$ and $KHCO_3$.

(A) Both are fairly soluble in water

(B) Both have similar hydrogen bonded structure of HCO_3^- ion in their solid

(C) Both solid consists of ion-dipole type of interaction

(D) Both on heating produces respective carbonate as residue

19. Which of the following is/are not correct for Lithium –

(A) It's hydrogen carbonate exist in solid form

(B) It's carbonate is most stable to heat amongest the group I-A metals

(C) It's main combustion product in excess oxygen is LiO_2

(D) It's chloride is soluble in pyridine

SECTION-II Integer : (Maximum Marks: 6)

Each question of this section have +3 marks.

1. How many reactions give correct products ?

(i) $2Na + NH_3\ (liq) \longrightarrow Na_2NH + H_2\uparrow$

(ii) $Na + (x+y)\ NH_3\ (g) \longrightarrow [Na(NH_3)_x]^{+1} + [e(NH_3)_y]^{-1}$

(iii) $Ni^{2+} + 4NH_3(excess) \longrightarrow \underset{\text{Blue solution}}{[Ni(NH_3)_4]^{2+}}$

(iv) $BaCl_2 + K_2CrO_4 \longrightarrow \underset{\text{soluble products}}{Ba^{2+} + CrO_4^{2-} + 2K^+ + Cl^-}$

(v) $NH_4Cl + NaNO_2 \xrightarrow{\Delta} NaCl + N_2O + H_2O$

(vi) $Hg_2Cl_2 + 2NH_3 \rightarrow \underset{\text{Nice black ppt}}{Hg + Hg\begin{matrix}NH_2\\Cl\end{matrix}} + NH_4Cl$

2. Find the sum of the oxidation state of S-atoms present in compounds of Lithophone.

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SECTION-I (i) Single Correct (Maximum Marks:45)

Each question of this section have +3 marks.

1. Which of the following process does not remove permanent hardness :

(A) Permutit process (B) Synthetic resins method

(C) Clark's method (D) Calgon's method

2. The radius of La^{+3} is 1.06 Å which of the following given values will be closest to the radius of Lu^{+3} (At. number of Lu = 71, La = 57)

(A) 1.6 Å (B) 1.4 Å

(C) 1.06Å (D) 0.85 Å

3. Which of the following does not evolve any gaseous product on heating :

(A) $NaNO_3$ (B) Na_2CO_3

(C) NH_4NO_3 (D) $(NH_4)_2CO_3$

4. Which of the following compounds are coloured in aq. solution

(A) $Ce(SO_4)_2$ (B) $TiCl_4$

(C) $K_3[Cu(CN)_4]$ (D) $ZnSO_4.7H_2O$

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5. Which of the following will act as best oxidizing agent.

(A) CrO_3 (B) MoO_3 (C) WO_3 (D) SO_3

6. Which of the following process is used to prepare H_2O_2 by :-

(A) Oxidation of 2-ethylanthraquinon by O_2

(B) Oxidation of 2-ethylanthraquinol by O_2

(C) Reaction of 2-ethylanthraquinol by H_2O

(D) Reduction of 2-ethylanthraquinol by H_2

7. In a sample of temporary hard water which of the following water softening process does not produced any precipitate of cation responsible for hardness?

(A) Boiling of water

(B) Addition of lime water

(C) Addition of sodium hexameta phosphate

(D) Addition of sodium carbonate

8. $[X] \underset{H^+}{\overset{OH^-}{\rightleftharpoons}} [Y]$

Both [X] and [Y] produce ink blue solution with H_2O_2 in presence of organic solvents and H_2SO_4, [X] & [Y] respectively

(A) MnO_4^-, MnO_4^{-2} (B) $Cr_2O_7^{-2}, CrO_4^{-2}$

(C) $CrO_4^{-2}, Cr_2O_7^{-2}$ (D) SO_3^{-2}, SO_4^{-2}

9. A colourless solid (X) on heating evolve CO_2 & gives white residue, which is soluble in water. Residue also give CO_2 on treatment with dil HCl.

The compound (X) is/are

(A) $LiHCO_3$ (B) $NaHCO_3$

(C) $CaCO_3$ (D) $Ca(HCO_3)_2$

10. Which of the following statement is **INCORRECT** for K_2MnO_4 ?

(A) It is prepared by MnO_2 in acidic medium

(B) It produce brown coloured compound on addition of water

(C) It produce $KMnO_4$ on addition of water

(D) It produce purple colour compound with O_3

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11. Which of the following is not correct about the chemistry of 3d and 4f series elements?

(A) 3d elements show more oxidation state than 4f series elements

(B) The energy difference between 3d and 4s orbital is very little.

(C) Europium (II) is more stable than cerium (II)

(D) The magnetic character in 3d series elements increases from scandium to copper

12. In which of the following reaction, H_2O_2 is acting as a reducing agent?

(A) $SO_2 + H_2O_2 \longrightarrow H_2SO_4$ (B) $2KI + H_2O_2 \longrightarrow 2KOH + I_2$

(C) $Ag_2O + H_2O_2 \longrightarrow 2Ag + H_2O + O_2$ (D) $PbS + 4H_2O_2 \longrightarrow PbSO_4 + 4H_2O$

13. The lowest degree of paramagnetism is shown by :

(A) $MnSO_4.4H_2O$ (B) $FeSO_4.6H_2O$

(C) $CuSO_4.5H_2O$ (D) $NiSO_4.6H_2O$

14. When chromite ore is heated with Na_2CO_3 strongly in air followed by cooling and washing with water, then brown residue is obtained which is of

(A) Fe_2O_3 (B) $Fe_2O_3.FeO$ (C) Na_2CrO_4 (D) $FeO.Cr_2O_3$

15. In portland cement, which of the following compounds are mainly present. (I) Dicalcium silicate. (II) Tricalcium silicate (III) Tricalcium aluminate

(A) (I) only
(B) (II) and (III) only
(C) (I) and (III) only
(D) I, II and III

SECTION-I (ii) Multiple Correct : (Maximum Marks: 12)

Each question of this section have +4 marks.

16. When H_2O_2 reacts with $Ce(SO_4)_2$:-

(A) Ce^{+4} is oxidised to higher oxidation state
(B) Ce^{+3} is produced
(C) O_2 is produced
(D) H_2 is evolved

17. HD gas is prepared by :-

(A) Reaction of D_2O with NaH
(B) Reaction of H_2O with NaD
(C) Electrolysis of D_2O
(D) Na_2O_2 with D_2O

18. Reaction which involve redox change is/are

(A) $Cu^{+2} + \overline{C}N_{(Excess)} \rightarrow$
(B) $FeS \xrightarrow{Roasting}$
(C) $CO_2 + MnO_4^- \xrightarrow{\text{weak alkaline solution}}$
(D) $Na_2CrO_4 + H_2SO_4 \rightarrow$

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SECTION-I (iii) Paragraph (SOT) : (Maximum Marks: 6)

Each question of this section have +3 marks.

Paragraph for Questions 19 and 20

Fajan's rule is very important for covalent character in ionic bond. According to this rule as the polarising power of cation and polarisibility of anion increases then covalent character in ionic bond increases.
Basis of Fajan's rule is also useful to predict various properties of ionic salt like nature of oxide, melting point, thermal stability and colour of ionic salt etc.

19. Select incorrect order of thermal stability.

(A) $Li_3N > Na_3N > K_3N$
(B) $LiCl < NaCl < KCl$
(C) $BeSO_4 < MgSO_4 < CaSO_4$
(D) $GeCl_4 > SnCl_4 > PbCl_4$

20. Select the correct order of basic strength.

(A) $CaO < ZnO$
(B) $ZnO < HgO$
(C) $PbO > Ag_2O$
(D) $Fe_2O_3 > FeO$

SECTION-I (i) Single Correct (Maximum Marks:57)

Each question of this section have +3 marks.

1. $Na[BH_4]$ is formed by reaction of :-

(A) Two electron deficient hydrides

(B) Two saline hydrides

(C) One saline and one electron deficient hydride

(D) One metallic and one electron deficinet hydride

2. Choose correct statement about various water softening method :

(A) In Colgon's method sodium hexametaphosphate ($Na_6P_6O_{18}$) is used

(B) In permutit method sodium aluminium silicate is used

(C) In synthetic resin method, all soluble minerals from water are removed.

(D) All of these

3. Choose correct order :

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(A) Solubility $Be(OH)_2 < Ba(OH)_2$

(B) Thermal stability LiOH < CsOH

(C) Magnetic moment $O_2 > O_2^-$

(D) All are correct

4. Which of the following is true about water :

(A) Hardness can be removed by passing through ion–exchange resin.

(B) Its presence can be detected by anhydrous $CuSO_4$

(C) It has maximum density at 4°C

(D) All of these

5. Out of Cobalt, Scandium, Iron and Zinc simple salts, which is attraced in a magnetic field?

(A) Zinc simple salts only

(B) Cobalt simple salts only

(C) Both Cobalt and Iron simple salts

(D) Both Scandium and Zinc simple salts

6. Among the VO_4^{3-}, CrO_4^{2-}, MnO_4^{2-} & FeO_4^{2-}, which of the following pair of oxoanion is strongest oxidant & weakest oxidant?

(A) VO_4^{3-}, CrO_4^{2-} (B) FeO_4^{2-}, VO_4^{3-}

(C) MnO_4^{2-}, FeO_4^{2-} (D) CrO_4^{2-}, MnO_4^{2-}

7. Fire extinguisher contain H_2SO_4 and :

(A) $BaCO_3$ (B) $NaHCO_3$ solution

(C) Na_2CO_3 (D) $CaCO_3$

8. When sulphur is heated with $NaOH_{(aq.)}$ the compounds formed are:

(A) $Na_2S + H_2O$ (B) $Na_2SO_3 + H_2O$

(C) $Na_2S + Na_2S_2O_3 + H_2O$ (D) $Na_2S_2O_3 + H_2O$

9. Among the following which has maximum solubility in water?

(A) KOH (B) CsOH

(C) LiOH (D) RbOH

10. When KCl is heated with conc. H_2SO_4 and solid $K_2Cr_2O_7$, we get :

(A) Chromyl chloride (B) Chromous chloride

(C) Chromic chloride (D) Chromic oxide

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11. Which sequence of reaction show chemical relation between sodium and its compound?

(A) $Na + O_2 \rightarrow Na_2O \xrightarrow{HCl_{(aq.)}} NaCl$
$Na_2O \xrightarrow{CO_2} Na_2CO_3 \xrightarrow{\Delta} Na$

(B) $Na + O_2 \rightarrow Na_2O \xrightarrow{H_2O} NaCl$
$Na_2O \xrightarrow{CO_2} Na_2CO_3 \xrightarrow{\Delta} Na$

(C) $Na + H_2O \rightarrow NaOH \xrightarrow{HCl} NaCl$
$NaOH \xrightarrow{CO_2} Na_2CO_3 \xrightarrow{\Delta} Na$

(D) $Na + H_2O \rightarrow NaOH \xrightarrow{CO_2} Na_2CO_3$
$NaOH \xrightarrow{HCl} NaCl \text{ (Molten)} \xrightarrow{\text{electrolysis}} Na$

12. For alkali metals, which one of the following trends is incorrect?

(A) Hydration energy : $Li^+ > Na^+ > K^+ > Rb^+$

(B) Ionization energy : Li > Na > K > Rb

(C) Density : Li < Na < K < Rb

(D) Atomic size : Li < Na < K < Rb

13. Hydrogen peroxide is now generally prepared on industrial scale by the :

(A) Action of H_2SO_4 on barium peroxide

(B) Action of H_2SO_4 on sodium peroxide

(C) Electrolysis of 50% H_2SO_4 & followed by hydrolysis

(D) Burning hydrogen in excess of oxygen

14. When silicon is boiled with caustic soda solution, the gas evolved is :

(A) O_2

(B) SiH_4

(C) H_2

(D) SiO_2

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15. Which of the following reactions does not form gaseous product?

(A) $PbO_2 + H_2O_2 \longrightarrow$

(B) $PbS + H_2O_2 \longrightarrow$

(C) Acidified $KMnO_4 + H_2O_2 \longrightarrow$

(D) $Cl_2 + H_2O_2 \longrightarrow$

16. The value of .....is less for D_2O as compared to H_2O.

(A) density (g mL^{-1}) at 20°C

(B) Boiling point

(C) Dielectric constant at 20°C

(D) Latent heat of vaporisation at 20°C

17. The tendency of 3d-metal ions to form stable complexes is due to their :

(A) Very low ionization energies

(B) Variable oxidation state

(C) Strong electronegative state

(D) High charge/size ratio and vacant d-orbital

18. The maximum number of unpaired electrons are in :

(A) Fe^{2+}

(B) Fe^{3+}

(C) Fe

(D) Fe^{4+}

19. The correct order of decreasing atomic radius is :-

(A) La > Ce > Eu > Yb

(B) La > Eu > Ce > Yb

(C) Yb > Eu > La > Ce

(D) Eu > La > Ce > Yb

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SECTION-I (ii) Multiple Correct : (Maximum Marks: 4)

Each question of this section have +4 marks.

20. Which of the following facts are the evidence for ionic nature of LiH.

(A) Molten LiH (m.p. 691°C) conducts electricity

(B) H_2 is liberated at the anode on electrolysis of the molten LiH.

(C) When LiH reacts with water liberates H_2

(D) Li gets deposited at cathod on electrolysis of molten LiH.

SECTION-I (i) Single Correct (Maximum Marks:24)

Each question of this section have +3 marks.

1. Which of the following sodium salt react with dil. H_2SO_4 & gives triatomic gas

(A) Na_2CO_3 (B) Na_2SO_3

(C) Na_2S (D) All of the above

2. In which of the following option N_2 gas is formed

(A) $Mg_3N_2 + H_2O \longrightarrow$ (B) $NH_4NO_3 \xrightarrow{\Delta}$

(C) $Zn + dil.HNO_3 \longrightarrow$ (D) $(NH_4)_2Cr_2O_7 \xrightarrow{\Delta}$

3. Which of the following salts of anion produce coloured vapours with conc. H_2SO_4

(A) NaCl (B) Na_2S (C) $Na_2S_2O_3$ (D) NaI

4. $3BCl_3 + 3NH_4Cl \xrightarrow{140^0C} B_3N_3H_3Cl_3 \xrightarrow{NaBH_4} B_3N_3H_6 \xrightarrow{3HCl} B_3N_3H_9Cl_3$.

The protonation in the last step occurs on :

(A) B-atom (B) N-atom

(C) Cl-atom (D) Can not be predicted

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5. Which of the following cation does not form soluble complex with excess of NH_3?

(A) Fe^{+3} (B) Zn^{+2} (C) Cu^{+2} (D) Ag^{+}

6. Which of the following element does not show –3 oxidation state.

(A) Bi (B) N (C) As (D) Cℓ

7. When H_2O_2 reacts with HIO_4 & NH_2OH separately, what are the iodine containing product & nitrogen containing product respectively.

(A) I^-, NH_3 (B) IO_3^-, NO_3^-

(C) IO_3^-, N_2O (D) I_2, N_2

8. Which of the following set is correct matching.

	Compound	Correct Match
(1)	CO $\longrightarrow$	Ethanol amine is correct absorber
(2)	H_2S $\longrightarrow$	H_2SO_4 (conc.) is correct drying agent
(3)	NH_3 $\longrightarrow$	Anhydrous $CaCl_2$ is correct drying agent
(4)	PH_3 $\longrightarrow$	$HgCl_2$ solution is correct absorber

SECTION-I (ii) Multiple Correct : (Maximum Marks: 28)

Each question of this section have +4 marks.

9. Which of the following oxy acid(s) on heating form one of the products in the lowest oxidation state of underlined atom when disproportionation reaction takes place.

(A) $H_3\underline{P}O_2$ (B) $H_3\underline{P}O_3$

(C) $H\underline{N}O_2$ (D) $H_2\underline{S}O_4$

10. In large scale NH_3 is prepared by Haber's process. Select the correct statement(s) for Haber's process

(A) ΔH_f^0 for NH_3 is –ve

(B) iron oxide with small amount $K_2O + Al_2O_3$ is used as catalyst

(C) high pressure favour the formation of NH_3

(D) Extremly high temperature favour the formation of NH_3 $\approx$2700°C

11. Which of the following statement(s) is/are correct?

(A) In Ostwald process for the preparation of HNO_3, Pt/Rh is used at catalyst.

(B) In Contact process for the preparation of H_2SO_4, V_2O_5 is used as catalyst.

(C) In Decan process for the preparation of Cl_2, $CuCl_2$ is used as catalyst.

(D) In Haber process for the preparation of NH_3, (iron oxide, Al_2O_3, K_2O) is used as catalyst.

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12. $NH_4C\ell O_4 \xrightarrow{\Delta} X_{(g)}$, (X) is produced by heating -

(A) $(NH_4)_2CO_3$

(B) $(NH_4)_2SO_4$

(C) $(NH_4)_2Cr_2O_7$

(D) NH_4NO_2

13. Consider the following statement(s) number of correct statement is/are

(A) Anhydride of HNO_3 is an acidic oxide

(B) $H_2S_2O_3$, $H_2S_2O_4$, $H_2S_2O_5$, $H_2S_xO_6$ all contain atleast one S-S linkage

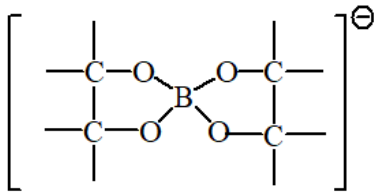
(C) Carbon monoxide is produced along with pure ZnO on heating basic Zinc Carbonate.

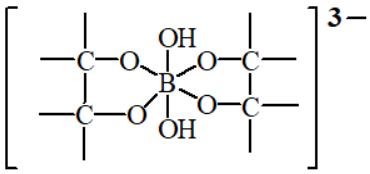
(D) All oxides of Sulphur and Chromium are acidic

14. On addition of Cis 1-2 diol in the aq. solution of boric acid -

(A) H^+ ion concentration increases

(B) H^+ ion concentration decreases

(C)  is formed

(D)  is formed

15. Which of the following substance is/are identified due to their unique characteristic smell.

(A) PH_3 (B) O_3

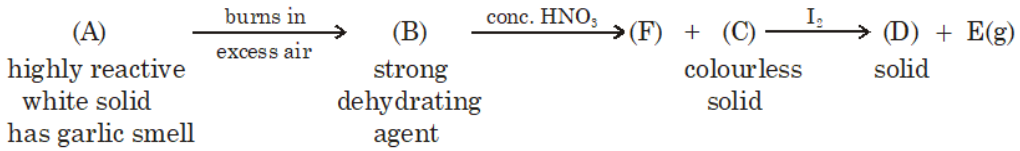
(C) NH_3 (D) H_2S

SECTION-I (iii) Paragraph (SOT) : (Maximum Marks: 6)

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Each question of this section have +3 marks.

Paragraph for Questions 16 and 17



16. When (C) reacts with NaCl, the state of hybridisation of N-atom in the products.

(A) sp^2 only (B) sp^2 and sp^3

(C) sp^3 only (D) sp and sp^2

17. Which of the following statements is/are incorrect

(A) 'E' produces HNO_2 and HNO_3 when reacts with water.

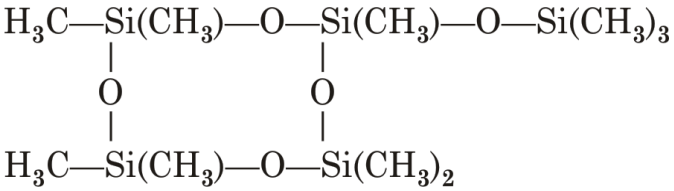
(B) 'D' is used to estimate CO gas

(C) 'B' has cage like structure

(D) E is diatomic gas

SECTION-III Integer : (Maximum Marks: 9)

Each question of this section have +3 marks.

1. 

Under hydrolytic condition, the number of the following compounds (R_nSiCl_{4-n}) which is/are used to prepare given silicon's compound(s).

$Si(CH_3)_4$, $(CH_3)_3SiCl$, $(CH_3)_2SiCl_2$, $(CH_3)SiCl_3$, $SiCl_4$

2. How many compounds liberate NH_3 on heating from the following.

$(NH_4)_2SO_4$ $(NH_4)_2CO_3$ NH_4Cl NH_4NO_3 NH_4NO_2

3. XeF_4 reacts with PF_5 as follows :

$XeF_4 + PF_5 \longrightarrow [XeF_3]^+ [PF_6]^-$

If total numbers of 90° angles in reactants = x and if total number of 90° angles in products = y, considering all effects of VSEPR theory, then calculate the value of (y –x).

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SECTION-I (i) Single Correct (Maximum Marks:30)

Each question of this section have +3 marks.

1. Hot conc. H_2SO_4 acts as moderately strong oxidising agent. It oxidises both metals and nonmetals. Which of the following element is oxidised by conc. H_2SO_4 into two gaseous products?

(A) Cu (B) C

(C) P (D) Zn

2. A black compound of manganese reacts with a halogen acid to give greenish yellow gas. When this gas reacts with excess NH_3 a non-inflammable gas is formed. In this process the oxidation state of nitrogen changes from _____.

(A) -3 to $+3$ (B) -3 to 0

(C) -3 to $+5$ (D) 0 to -3

3. In the preparation of compounds of Xe, Bartlett had taken $O_2^+ Pt F_6^-$ as a base compound. This is because

(A) both O_2 and Xe have same size.

(B) both O_2 and Xe have same electron gain enthalpy.

(C) both O_2 and Xe have almost same ionisation enthalpy.

(D) both Xe and O_2 are gases.

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4. In which of the following option N_2O gas is formed

(A) $P_4 + HNO_3 \longrightarrow$ (B) $NH_4SO_4 \xrightarrow{\Delta}$

(C) $Zn + dil.HNO_3 \longrightarrow$ (D) $NaNO_2 + NH_4Cl \longrightarrow$

5. Metal 'M' + white phosphorus $\longrightarrow$ 'X'

'X' $\xrightarrow{H_2O}$ Z + 'Y'

(gas)

$CuSO_4 + Y \longrightarrow$ black ppt.

(aq.)

Then choose correct statement about gas 'Y'

(A) Gas 'Y' is more basic than NH_3

(B) Bond angle of 'Y' > NH_3

(C) 'Y' has rotten fish smell

(D) Does not obey Lewis octet rule

6. $O_3 + NO_2 \longrightarrow O_2 + \text{'P'}$
(Oxide of nitrogen)
In above reaction NO_2 act as reducing agent then correct statement about 'P' is :-

(A) All N–O bond lengths are identical

(B) All $\widehat{ONO}$ angles are identical

(C) 'P' is anhydride of HNO_3

(D) Oxidation state of nitrogen in 'P' is +3

7. Product in which oxidation number of nitrogen is non-zero -

(A) $(NH_4)_2\,Cr_2O_7 \xrightarrow{\Delta}$ product

(B) $(NH_4)NO_2 \xrightarrow{\Delta}$ product

(C) $NH_4NO_3 \xrightarrow{\Delta}$ product

(D) All of the above

8. Which of the following metal is inert towards dil.HNO_3 (20%)

(A) Au (B) Cu

(C) Zn (D) Ag

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9. An atom have three electrons in outermost shell and $ns^2np^6nd^{10}$ configuration in penultimate shell. Select the incorrect about that atom -

(A) p-block element (B) 3rd period element

(C) 13th group element (D) Boron family element

10. Select correct order about the magnitude of energy which is associated with given change:
$X(g) + e^- \rightarrow X^-(g)$ (X = halogen atom)

(A) F > Cl > Br > I (B) Cl > F > Br > I

(C) Cl > Br > F > I (D) F > Br > Cl > I

SECTION-I (ii) Multiple Correct : (Maximum Marks: 12)

Each question of this section have +4 marks.

11. Which of the following compound is/are produced during heating borax ($Na_2B_4O_7 \cdot 10H_2O$)

(A) $Na_2B_4O_7$(Anhydrous)

(B) $NaBO_3$

(C) $NaBO_2$

(D) B_2O_3

12. Select the correct statement about V group hydrides [NH_3, PH_3, SbH_3, BiH_3]

(A) The thermal stability of hydrides decreases from NH_3 to BiH_3

(B) The reducing character of hydrides increases from NH_3 to BiH_3

(C) Bond angle decreases from NH_3 to BiH_3

(D) Basicity also decreases from NH_3 to BiH_3

13. When PCl_3 undergoes in hydrolysis then oxyacid of phosphorus X is formed then select the correct statements about X.

(A) Two P-OH bond

(B) one P-H bond

(C) one P = O bond

(D) Three P-OH bond

SECTION-I (iii) Paragraph (SOT) : (Maximum Marks: 12)

Each question of this section have +3 marks.

Paragraph for Questions 14 and 15

NaOH is one of the most important reagent :

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14. Which of the following element does not disproportinate when treated with NaOH :

(A) Br_2

(B) P_4

(C) S

(D) Na

15. Which of the following does not dissolve in excess of NaOH :

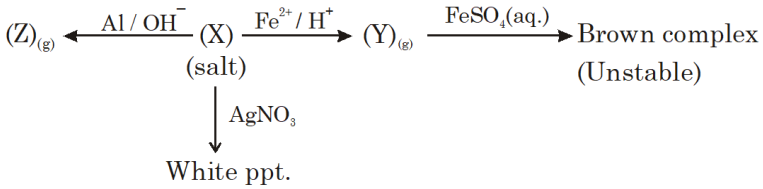
(A) $FeCl_3$

(B) $AlCl_3$

(C) $CrCl_3$

(D) $2PbCO_3.Pb(OH)_2$

Paragraph for Questions 16 and 17



16. Identify the 'Y' & 'Z'

(A) NO_2, NH_3 (B) NO, NH_3

(C) N_2O, NO_2 (D) NO, H_2

17. What is the oxidation no. of central metal in brown colour complex -

(A) +2 (B) +3

(C) +1 (D) zero

SECTION-I (iv) Matching List : (Maximum Marks: 3)

Each question of this section have +3 marks.

18.

	List-I (Reaction)		List-II (Gas evolved)
(P)	Sn + conc. HNO_3	(1)	H_2S
(Q)	Zn + dil.H_2SO_4	(2)	NH_3
(R)	S^{2-} + dil. HCl	(3)	NO_2
(S)	NO_2^- + Zn + NaOH	(4)	H_2

(A) P ⟶ 2 ; Q ⟶ 3 ; R ⟶ 4 ; S ⟶ 1

(B) P ⟶ 3 ; Q ⟶ 4 ; R ⟶ 2 ; S ⟶ 1

(C) P ⟶ 2 ; Q ⟶ 4 ; R ⟶ 3 ; S ⟶ 1

(D) P ⟶ 3 ; Q ⟶ 4 ; R ⟶ 1 ; S ⟶ 2

SECTION-II Integer : (Maximum Marks: 6)

Each question of this section have +3 marks.

1. $(CH_3)_nSiCl_{4-n}$ are called methyl subtituted chlorosilanes. If n = one, find maximum number of Si–O linkages on each Si in product obtained by hydrolysis & condensation.

2. $(HPO_3)_n \xrightarrow{+xH_2O}$ 'Y' (Poly phosphoric acid)

If only 'Y' is from in above reaction and 'Y' have three P–O–P linkage then value of (x) will be :-

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SECTION-I (i) Single Correct (Maximum Marks:30)

Each question of this section have +3 marks.

1. Which of the following set of four quantum numbers is possible?

(A) $n = 4, \ell = 3, m = 2, s = +\frac{1}{2}$

(B) $n = 3, \ell = 3, m = 0, s = +\frac{1}{2}$

(C) $n = 2, \ell = 1, m = -2, s = +\frac{1}{2}$

(D) $n = 5, \ell = 3, m = +4, s = +\frac{1}{2}$

2. Which of the following order for given property is wrong –

(A) $O^{2-} > O > O^{+}$ (size)

(B) Mg < S < Mn (valence electron)

(C) $Fe^{2+} < Mn^{2+} < Cr^{3+}$ (total number of unpaired electron)

(D) Sc < Y < La (atomic size)

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3. The order of which of the following oxides is arranged according to decreasing basic nature ?

(A) Na_2O, MgO, Al_2O_3

(B) Al_2O_3, Na_2O , MgO

(C) Al_2O_3, MgO, Na_2O

(D) MgO, Na_2O, Al_2O_3

4. Select the correct radii order –

(A) K > Ca > P > Mg > N

(B) $Cl^-_{(g)} > S^{2-}_{(g)} > F^-_{(g)} > Na^+_{(g)}$

(C) $Be^{2+}_{(aq)} > Li^+_{(aq)}$

(D) Co ≃ Ni ≃ Zn

5. Select the correct ionisation energy order.

(A) Ag < Cu < Au

(B) O < N < F

(C) La < Y < Sc

(D) All of these

6. Which of the following conversion is exothermic?

(A) $Na \longrightarrow Na^+$

(B) $Li \longrightarrow Li^-$

(C) $N \longrightarrow N^-$

(D) $Mg \longrightarrow Mg^-$

7. The first seven successive ionisation energy (in eV/atom) of element 'X' are 8 eV/atom, 16 eV/atom, 33 eV/atom, 45 eV/atom, 166 eV/atom, 205 eV/atom and 246 eV/atom respectively. The element 'X' can be

(A) C (B) O (C) N (D) Si

8. Lithium acts as best reducing agent in aqueous medium because
Reason-1 : Lithium has high sublimation energy.
Reason-2 : Lithium has large I.E. among group 1 metals.
Reason-3 : Li^+ ion has highest hydration energy among group 1 metal ions.
Which of the above reason is correct.

(A) Reason 1 alone (B) Reason 1 and 3 both

(C) Reason 3 alone (D) No reason is correct

9. Which of the following order is incorrect.

(A) $C\ell^- < K^+ < Ca^{+2}$ (I.E.) (B) Pd < Ag (I. E.)

(C) $Li^+ > Na^+$ (E.A.) (D) Mg^{+2} (aq) < Ca^{+2} (aq) (ionic mobility)

10. Which of the following is not a result of d-contraction or f-contraction?

(A) Greater I.E. of Ga compared to Aℓ.

(B) Smaller size of Ga compared to Aℓ.

(C) Hg has highest I.E. among : Zn, Cd, Hg

(D) Exceptional configuration of Lanthanum.

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SECTION-I (ii) Multiple Correct : (Maximum Marks: 20)

Each question of this section have +4 marks.

11. Which of the following is/are the incorrect match for atom of elements?

(A) [Rn] $6d^2\ 7s^2$ ⟶ d-block, 4th group

(B) [Xe] $4f^{14}\ 5d^1\ 6s^2$ ⟶ d-block, 3rd group

(C) [Xe] $4f^7\ 6s^2$ ⟶ f-block, 9th group

(D) [Kr] $4d^{10}$ ⟶ d-block, 12th group

12. Choose the correct statement(s) :

(A) Smallest anion in the periodic table in H^-

(B) Ionic mobility of $Mg^{2+}_{(aq)}$ is greater than that of $Be^{2+}_{(aq)}$

(C) Electronegativity of Pb^{2+} is more than that of Pb^{4+}

(D) Second ionisation of Al is higher than second ionisation of Mg.

13. The correct order of electron affinity of the element or ion is/are?

(A) O > Se (B) $F^+ > Cl^+$ (C) P > N (D) Ne > Ar

14. Which of the following oxide(s) will react with H_2SO_4 as well as KOH solution to produce salt?

(A) Al_2O_3 (B) BaO (C) SnO (D) ZnO

15. If an element 'X' belongs from 5th period and 10th group then select the correct statement regarding element X.

(A) Last electron of element 'X' enters in n = 4 and $\ell = 2$

(B) It is a diamagnetic specie

(C) It is not a transition element

(D) It is a coinage metal

SECTION-I (iii) Matching List : (Maximum Marks: 3)

Each question of this section have +3 marks.

16. Match list-I with list-II

	List-I (Periodic properties)		List-II (Order)
(P)	Ionisation energy	(1)	Sn < Pb
(Q)	Radii	(2)	N < C < O
(R)	Electronegativity	(3)	Li < Be < B
(S)	Electron affinity	(4)	$O^- < S^-$

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(A) P ⟶ 4;Q ⟶ 1;R ⟶ 3;S ⟶ 2 (B) P ⟶ 1;Q ⟶ 2;R ⟶ 3;S ⟶ 4

(C) P ⟶ 4;Q ⟶ 3;R ⟶ 2;S ⟶ 1 (D) P ⟶ 1;Q ⟶ 4;R ⟶ 2;S ⟶ 3

SECTION-II Integer : (Maximum Marks: 9)

Each question of this section have +3 marks.

1. From the following list of enthalpies, find the number of enthalpies which are not required to prepare Born-Haber cycle of NaI

ΔH_{Sub} (Na), ΔH_{Sub} (I_2), ΔH_{Diss} (I_2), ΔH_{Hyd} (Na^+),
ΔH_{Hyd} (I^-), $\Delta H_{I.E.}$ (I), ΔH_{eg1} (I), $\Delta H_{L.E.}$ (NaI),
$\Delta H_{I.E.}$ (Na^+)

2. In how many following pairs, first species has higher ionisation energy than second species :
(i) Na^+, Mg^{2+} (ii) S, Cl (iii) Cu, Zn (iv) Xe, Kr
(v) B, Be (vi) O^{2-}, O (vii) Al, Si (viii) Cl^-, Cl

3. In aqueous solution, how many following species have higher hydration energy than Li^+.
Na^+, K^+, Cl^-, F^-, Rb^{+1}, Be^{2+}

SECTION-IV Match the Column : (Maximum Marks: 16)

Each question of this section have +8 marks
(Partial marks of each correct is +2)

1.

	Column-I		Column-II
(A)	$\Delta H < O$ (Exothermic)	(P)	$N^+(g) \longrightarrow N^{+2}(g)$
(B)	$\Delta H > O$ (Endothermic)	(Q)	$N(g) \longrightarrow N^-(g)$
(C)	$\Delta H_{IE_2}(N)$	(R)	$N(g) \longrightarrow N^{+2}(g)$
(D)	$\Delta H_{Eg_1}(N)$	(S)	$N^-(g) \longrightarrow N^{-2}(g)$
		(T)	$F^-(g) \longrightarrow F^-(aq)$

2.

	Column-I (Ionic Compound)		Column-II (Properties)
(A)	Li_2O	(P)	Highest lattice energy among given compounds
(B)	Na_2O	(Q)	Basic oxide
(C)	MgO	(R)	Amphoteric oxide
(D)	$A\ell_2O_3$	(S)	Least lattice energy among given compounds
		(T)	The gaseous cation (among given ionic ompounds) will be most extensively hydrated in aqueous medium

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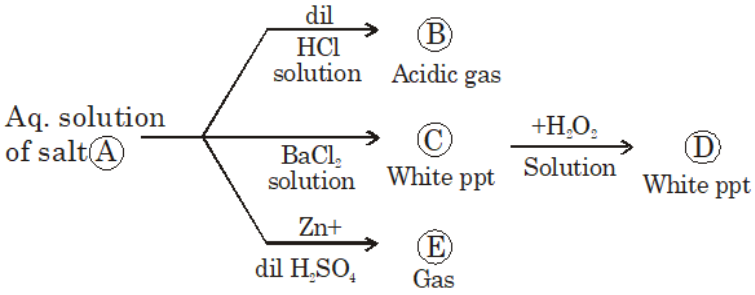
SECTION-I (i) Single Correct (Maximum Marks:33)

Each question of this section have +3 marks.

1. Which of the following reagent can distinguish between CO_2 and SO_2 gas ?

(A) Hydrogen peroxide solution
(B) Chlorine water
(C) Potassium dichromate solution
(D) Both (B) and (C)

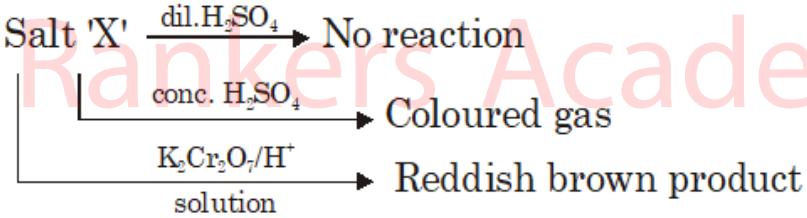
2.



The gas (E) is :-

(A) CO_2
(B) SO_2
(C) H_2S
(D) NO_2

3.



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Salt 'X' may contain anion :-

(A) Cl^-
(B) Br^-
(C) NO_2^-
(D) F^-

4. Which of the following ion does not convert yellow solution of $FeCl_3$ to green solution of $FeCl_2$ as a final product?

(A) $S_2O_3^{2-}$
(B) S^{2-}
(C) Sn^{+2}
(D) CH_3COO^-

5. In which of the following process yellow ppt is not obtained.

(A) Lead acetate solution is treated with K_2CrO_4
(B) $AgNO_3$ solution is treated with K_2CrO_4
(C) $AgNO_3$ solution is treated with KI
(D) H_2S is passed through solution of $CdSO_4$

6. Which of the following anion(s) can decolourise Br_2 water

(A) S^{-2}
(B) SO_3^{-2}
(C) NO_2^-
(D) All

7. $(X)_{\text{Salt}} \xrightarrow{Zn/OH^{\ominus}} (Y)_{(g)} \xrightarrow{dil.HCl}$ white fumes

$(Y)_{(g)}$ | red litmus → Blue

Where 'X' & 'Y' are respectively.

(A) $NaNO_2$ & NO

(B) $NaNO_3$ & NH_3

(C) Na_2S & H_2S

(D) $Na_2S_2O_3$ & SO_2

8. An aqueous solution of sodium salt gives pale yellow ppt and a gas with pungent odour on reaction with warm dilute hydrochloric acid then anionic part of salt is:

(A) Sulphate

(B) Sulphide

(C) Thiosulphate

(D) Sulphite

9. Which of the following anion does not produce ppt with $BaCl_2$ solution however gives ppt with $AgNO_3$ solution.

(A) $CO_3^{2-}(aq.)$

(B) $C_2O_4^{2-}(aq.)$

(C) $NO_3^{-}(aq.)$

(D) $S^{2-}(aq.)$

10. Consider following two observation in test tubes contain clear solution.

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Test tube-1 | $Ca(NO_3)_2$ → No ppt. $\xrightarrow{Pb(NO_3)_2}$ white ppt. $\xrightarrow[S_2O_3^{-2}]{\text{excess of}}$ soluble

Test tube-2 | $\underset{\text{excess}}{Ca(NO_3)_2}$ → No ppt. $\xrightarrow{\text{Boil}}$ white ppt. $\xrightarrow[\text{with dil. HCl}]{\text{Aq. suspension}}$ $SO_2\uparrow$

Select statement which is may be true?

(A) Test tube–1 contain CO_3^{-2} ions

(B) Test tube–2 contain CO_3^{-2} ions

(C) Test tube–1 contain $S_2O_3^{-2}$ ions

(D) Test tube–2 contain $S_2O_3^{-2}$ ions

11. $\underset{\text{Salt}}{(X)} + Na_2[Fe(CN)_5\overset{\oplus}{N}O] \xrightarrow{\text{basic solution}}$ Purple Colour

Which of the following statement is correct for purple colour complex?

(A) Denticity of new ligand formed is 3

(B) It's magnetic nature is paramagnetic

(C) It is low spin complex

(D) Hybridisation of Fe is sp^3d^2

SECTION-I (ii) Multiple Correct : (Maximum Marks: 4)

Each question of this section have +4 marks.

12. Colour of iodine solution is disappeared by shaking it with aqueous solution of :-

(A) $S_2O_3^{2-}$ (B) $NO_2^{\ominus}$

(C) CO_3^{2-} (D) Cl_2

SECTION-I (iii) Paragraph (SOT) : (Maximum Marks: 12)

Each question of this section have +3 marks.

Paragraph for Questions 13 and 14

Qualitative analysis of inorganic salts means the identification of cations and anions present in the salt or a mixture of salts. Inorganic salts may be obtained by complete or partial neutralisation of acid with base or vice-versa. In the formation of a salt, the part contributed by the acid is called anion and the part contributed by the base is called cation.

13. An acidic radical when treated with conc. sulphuric acid and heated strongly, form coloured vapour/gas. When $AgNO_3$ solution is added to aqueous solution of acidic radical no precipitate is formed. Identify the acidic radical.

(A) NO_3^- (B) NO_2^- (C) Br^- (D) I^-

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14. Which of the following interfering radical does not form volatile product on addition of conc. H_2SO_4 followed by heating.

(A) PO_4^{3-} (B) F^- (C) $C_2O_4^{2-}$ (D) BO_3^{3-}

Paragraph for Questions 15 and 16

Qualitative analysis of inorganic salt is carried out through the reactions which are easily perceptible to our senses such as sight and smell. Such reactions involve:

(a) Formation of a precipitate

(b) Change in colour

(c) Evolution of gas etc.

15. Which of the following acidic radical when treated with sulphanilic acid in the presence of dil. acetic acid followed by the reaction with 1-nepthyl amine red dye is formed. When the above acidic radical treated with conc. H_2SO_4 as brown gas is evolved.

(A) NO_2^- (B) NO_3^- (C) I^- (D) SO_3^{2-}

16. Which of the following is the specific test of sulphide

(A) Methylene blue test (B) layer test

(C) Chromyl chloride test (D) Brown ring test

SECTION-I (iv) Matching List : (Maximum Marks: 3)

Each question of this section have +3 marks.

17. Match the following list :-

	List - I (Reaction)		List - II (Characteristic odour/product)
(P)	$NO_2^{\ominus} \xrightarrow{\text{conc. HCl}}$	(A)	Coloured odd $e^{\ominus}$ species
(Q)	$S_2O_3^{-2} \xrightarrow{\text{dil. } H_2SO_4}$	(B)	Vinegar smell
(R)	$HCO_3^{\ominus} \xrightarrow{\text{dil. } H_2SO_4}$	(C)	Yellowish-white turbidly
(S)	$CH_3COO^{\ominus} \xrightarrow{\text{dil. HCl}}$	(D)	Colourless gas comes out with brisk effervescence

(A) P ⟶ 4;Q ⟶ 3;R ⟶ 1;S ⟶ 2

(B) P ⟶ 2;Q ⟶ 4;R ⟶ 1;S ⟶ 3

(C) P ⟶ 1;Q ⟶ 3;R ⟶ 4;S ⟶ 2

(D) P ⟶ 3;Q ⟶ 1;R ⟶ 4;S ⟶ 2

SECTION-II Integer : (Maximum Marks: 9)

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Each question of this section have +3 marks.

1. Among the following, total number of acidic radicals gives colourless volatile gas/vapour with dilute H_2SO_4 is.
$C_2O_4^{2\ominus}$, $Cl^{\ominus}$, $Br^{\ominus}$, $NO_2^{\ominus}$, $NO_3^{\ominus}$, $CH_3COO^{\ominus}$, $CO_3^{2\ominus}$, $SO_3^{2\ominus}$

2. Find number of species which produces ppt/turbidity in lime water
$CO_{2(g)}$, $HCO_{3(aq.)}^{-}$, $CO_{3(aq.)}^{2-}$, $HSO_{3(aq.)}^{-}$, $SO_{3(aq.)}^{2-}$, $HCl_{(g)}$, $SO_{2(g)}$

3. Find the number of gas(es)/vapour(s) changed the purple colour of acidified $KMnO_4$.
CO_2, SO_2, NO_2, H_2S, HCl, CH_3COOH

SECTION-I (i) Single Correct (Maximum Marks:24)

Each question of this section have +3 marks

1. Metals which are not dissolved in conc. HNO_3 due to formation of passive film of oxide.

(A) Hg, Cu (B) Ag, Zn

(C) Au, Pt (D) Al, Cr

2. Which of the following metal cation gives green colour in the borax bead test in oxidizing flame in cold condition.

(A) Cu (B) Co

(C) Cr (D) Ni

3. Which of the following salt produce coloured gas with dil.H_2SO_4 ?

(A) NaBr (B) $NaNO_2$

(C) $NaNO_3$ (D) NaI

4. Oxidation number of silver is NOT changed when –

(A) $Ag_2SO_{3\,(Aq)}$ is boiled

(B) Ag_2CO_3 crystal is strongly heated

(C) $AgNO_3$ reacts with aqueous $[Co(NH_3)_6]Cl_3$ solution

(D) $AgNO_{3\,(Aq.)}$ solution reacts with Cu-metal

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5. Which of the following cation produce white coloured ppt. with dil. HCl :-

(A) Hg_2^{2+} (B) Pb^{2+}

(C) Ag^+ (D) All of these

6. If in sodium salt 'A' anion have oxidation state of central atom is +3. Salt 'A' react with conc. H_2SO_4 and produce two gaseous product (no one is SO_2). Select the correct alternative about salt 'A' :-

(A) $Salt_{(aq.)}$ 'A' + $KMnO_4/H^+ \xrightarrow[\Delta]{60^\circ}$ brown ppt.

(B) $Salt_{(aq.)}$ 'A' + $\underset{\text{neutral sol.}}{FeCl_3} \longrightarrow$ deep red colour sol.

(C) $Salt_{(aq.)}$ 'A' + dil.$H_2SO_4 \longrightarrow$ no observable change

(D) Salt 'A' contains CH_3COO^- anion

7. Select the incorrect match :-

	Test of cations	Group Reagent
(1)	Cu^{+2}, Cd^{+2}, Hg^{+2}	H_2S with dil. HCl
(2)	Fe^{+3}, Cr^{+3}, $A\ell^{+3}$	$NH_4OH + NH_4Cl$
(3)	Ba^{+2}, Sr^{+2}, Ca^{+2}	$(NH_4)_2CO_3 + NH_4Cl$
(4)	Mn^{+2}, Zn^{+2}, CO^{+2}	H_2S in acidic medium

8.

$$\underset{\text{salt}}{(X)} + Zn \text{ powder} \xrightarrow[\Delta]{OH^{\ominus}} 'Y'_{(g)} \xrightarrow{\text{dil. HCl}} \text{white fumes}$$

$$(X) \xrightarrow{\text{sulphamic acid}} Z_{(g)}$$

Identify 'Y' & 'Z' respectively

(A) SO_2 , H_2S (B) N_2O, NH_3

(C) NH_3, N_2 (D) N_2, N_2O

SECTION-I (ii) Multiple Correct : (Maximum Marks: 12)

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Each question of this section have +4 marks

9. Which of the following compound is/are produced during heating borax ($Na_2B_4O_7 \cdot 10H_2O$)

(A) $Na_2B_4O_7$(Anhydrous) (B) $NaBO_3$

(C) $NaBO_2$ (D) B_2O_3

10. $Na_2S_2O_3$(solution) $\xrightarrow{'X'/H^+}$ coloured solution

Where 'X' is/are -

(A) $CuSO_4$(aq.) (B) $FeCl_3$(aq.)

(C) Cl_2 water (D) $Cr_2O_7^{-2}$

11. Aqueous solution containing Al^{+3} , Mg^{+2}, Fe^{+3}, Ni^{+2} is treated with NH_4Cl & NH_4OH. Which ion(s) is/are precipitated

(A) Al^{+3} (B) Mg^{+2}

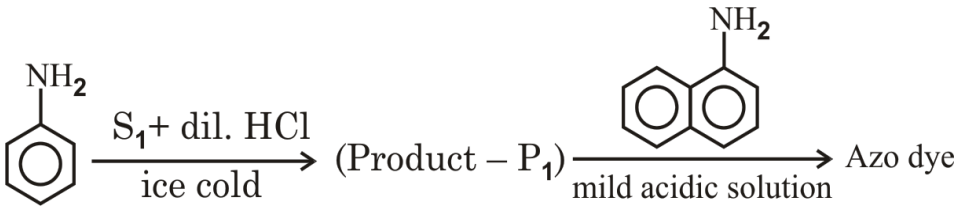
(C) Fe^{+3} (D) Ni^{+2}

SECTION-I (iii) Paragraph (SOT) : (Maximum Marks: 6)

Each question of this section have +3 marks

Paragraph for Questions 12 and 13

Consider two potassium salt (S_1) and (S_2) which gives following observation



$$CuSO_{4(aq.)} \xrightarrow{Salt - S_2} (Product - P_2)\downarrow + K_2SO_4 + \text{poisonous gas (G)}$$

(Product – P_1) + (Product – P_2) ⟶ Show Sandmeyer's reaction

12. Salt–S_2 is

(A) KCN

(B) KBr

(C) KNO_2

(D) NaCN

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13. Select correct for anion in salt S_1

(A) can react with $KMnO_4/H^+$

(B) react with KI/H^+

(C) gives NH_3 with KOH + Al

(D) All are correct

SECTION-II Integer : (Maximum Marks: 12)

Each question of this section have +3 marks

1. Find the total number of oxidising agents which can oxidise SO_2 to SO_4^{-2} in acidic medium.
Fe^{+2} , $Cr_2O_7^{-2}$, MnO_4^- , Br_2 water , H_2S , KIO_3 , Cl_2 water , H_2O_2

2. How many of the following salt on heating in bunsen flame gives characteristic green flame:-
$BaCl_2$, $CaCl_2$, LiCl , NaCl , $MgCl_2$

3. $Na_2CrO_{4(Aq.)} \xrightarrow{conc.\ H_2SO_4}$ product
What is the difference between oxidation number of Cr in product and reactant :-

4. Find the number of yellow colour sulphide(s) in the following :-
CdS, Sb_2S_3, As_2S_3, As_2S_5, SnS_2, ZnS, PbS

SECTION-IV Match the Column : (Maximum Marks: 16)

Each question of this section have +8 marks
(Partial marks of each correct is +2)

1.

	Column I		Column II
(A)	$S_2O_3^{2-}$	(P)	give fumes/gas with dil H_2SO_4
(B)	$NO_3^{\ominus}$	(Q)	give violet coloured ppt. with $[Ni(en)_3](NO_3)_2$
(C)	$NO_2^{\ominus}$	(R)	give white ppt with $BaCl_2$ solution
(D)	S^{2-}	(S)	identified by brown ring test
		(T)	identified by methylene blue dye test

2. Match the folloiwng column :-

	Column - I (Test of cation/anion)		Column - II (Observation)
(A)	NO_2^- + dil. H_2SO_4 + $FeSO_4$(Aq.)	(P)	Purple colour solution
(B)	S^{-2} + $Na_2[Fe(CN)_5NO]$	(Q)	Brown ring formation
(C)	Flame test of Ba^{2+}	(R)	Colourless solution
(D)	Pb^{+2} + NaOH (excess)	(S)	Apple green colour flame
		(T)	Complex compound is formed in product side

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SECTION-I (i) Single Correct (Maximum Marks:24)

Each question of this section have +3 marks.

1. Manganese salt + PbO_2 + conc. HNO_3 ⟶ the solution acquires purple colour. The colour is due to :-

(A) $HMnO_4$ (B) $Pb(NO_3)_2$

(C) $Mn(NO_3)_2$ (D) MnO_2

2. When borax is heated on platinum loop wire forms a glass like bead which is made-up of:-

(A) Sodium tetraborate

(B) Sodium metaborate

(C) Sodium metaborate and boric anhydride

(D) Boric anhydride and sodium tetraborate

3. A light green coloured salt soluble in water gives black precipitate on reaction with Na_2S, which dissolves readily in HCl. the metal ion present is:-

(A) Co^{2+} (B) Fe^{2+}

(C) Ni^{2+} (D) Mn^{2+}

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4. Which metal gives blue ash when its salt is heated with Na_2CO_3 solid and $Co(NO_3)_2$ on a charcoal piece?

(A) Cu (B) Mg

(C) Al (D) Zn

5. Cl_2 gas can not be prepared by –

(A) Electrolysis of fused NaCl (B) Acidification of Bleaching powder

(C) Reaction of conc. HCl with $KMnO_{4(aq.)}$ (D) Heating KCl with conc. H_2SO_4

6. Acidic aqueous solution contain $CuSO_4$, $ZnSO_4$, $MgSO_4$, $CdSO_4$ when react with $H_2S(g)$, Ions which is/are precepitated as metal sulphide.

(A) Cu^{+2}, Cd^{+2} (B) only Cu^{+2}

(C) Mg^{+2}, Zn^{+2} (D) Cu^{+2}, Zn^{+2}, Cd^{+2}

7. Select correct match

(A) Ag_2CrO_4 – Yellow ppt. (B) $[Fe(SCN)_3]$ – Green ppt.

(C) CdS – Black ppt. (D) $3HgO.HgCO_3$ – Reddish brown ppt.

8. Which of the following salt is not soluble in HNO_3 (1 : 1) but soluble in aquaregia?

(A) PbS (B) Ag_2S

(C) HgS (D) Ag_2CO_3

SECTION-I (ii) Multiple Correct : (Maximum Marks: 24)

Each question of this section have +4 marks.

9. Which of the following cation(s) will give precipitate with reagent NH_4Cl and NH_4OH :

(A) Ca^{2+} (B) Al^{3+}

(C) Bi^{3+} (D) Mg^{2+}

10. Which of the following reagent will give test with Fe^{3+} ions?

(A) NH_4SCN (B) $K_3[Fe(CN)_6]$

(C) NH_4OH (D) $K_4[Fe(CN)_6]$

11. Which of the following metal cation(s) gives white ppt. with Na_2CO_3?

(A) Fe^{3+} (B) Mg^{2+}

(C) Cr^{3+} (D) Zn^{2+}

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12. Select reaction in which precipitate is formed?

(A) $Ag^+ + SO^{-2}_{3(excess)} \rightarrow$

(B) $Ba^{2}_{(aq.)} + S_2O_3^{2-} \rightarrow$

(C) $Cu^{2}_{(aq.)} + S_2O_3^{2-} \rightarrow$

(D) $Hg^{+2}_{(aq.)} + S^{2-} \rightarrow$

13. Select the following reaction(s) in which chemical composition of product precipitate is changed when exposed to air for some time :-

(A) $BaCl_2(aq.) + Na_2SO_3\ (aq.) \longrightarrow$ ppt - 1

(B) $FeSO_4\ (aq.) + NaOH\ (aq.) \longrightarrow$ ppt - 2

(C) $BaS + ZnSO_4\ (aq.) \longrightarrow$ ppt - 3

(D) $MnSO_4\ (aq.) + NaOH\ (aq.) \longrightarrow$ ppt - 4

14. A test tube contain aqueous solution of $CuSO_4$ which can precipitate metal ion –

(A) $H_2S_{(g)}$ (B) $HCl_{(g)}$

(C) $NH_{3(g)}$ (excess) (D) $PH_{3(g)}$

SECTION-I (iii) Paragraph (SOT) : (Maximum Marks: 12)

Each question of this section have +3 marks.

Paragraph for Questions 15 and 16

In the group analysis of cations, group reagents are used to precipitate out cations.

15. Cations of which group are precipitated in the form of sulphides :-

(A) Group II and III

(B) Group II and IV

(C) Group II and V

(D) Group VI and III

16. During the analysis of group III cation, NH_4OH is added in presence of NH_4Cl. Which of the following factor has no significance for the addition of NH_4Cl :-

(A) NH_4Cl decreases the concentration of OH^- by common ion effect

(B) NH_4Cl form buffer solution with NH_4OH

(C) Concentration of NH_4^+ ion is decreased by the NH_4Cl addition

(D) All of these

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Paragraph for Questions 17 and 18

The basic radical are detected by various reagent through precipitation reaction and some of the precipitate soluble in excess of reagent.

17. Which of the following cation does not form precipitate on reaction with KI.

(A) Pb^{+2}

(B) Cu^{+2}

(C) Bi^{+3}

(D) Zn^{+2}

18. Which of the following cation does not form soluble complex with excess of NaOH?

(A) Cr^{+3}

(B) Zn^{+2}

(C) Cu^{+2}

(D) Sn^{+2}

SECTION-I (iv) Matching List : (Maximum Marks: 3)

Each question of this section have +3 marks.

19. Match the folloiwng list :-

	List - I (Compounds)		List - II (Reagents)
(P)	$PbCl_2$	(1)	Insoluble in water but soluble in NH_3 solution
(Q)	AgCl	(2)	Insoluble in cold water but soluble in hot water
(R)	$FeCl_3$(Neutral solution)	(3)	Produces prussian blue on reaction with $K_4[Fe(CN)_6]$
(S)	$ZnCl_2$	(4)	Soluble in excess of $NaOH_{(aq.)}$

(A) P $\longrightarrow$ 2;Q $\longrightarrow$ 1;R $\longrightarrow$ 3;S $\longrightarrow$ 4

(B) P $\longrightarrow$ 2;Q $\longrightarrow$ 1;R $\longrightarrow$ 4;S $\longrightarrow$ 3

(C) P $\longrightarrow$ 3;Q $\longrightarrow$ 4;R $\longrightarrow$ 1;S $\longrightarrow$ 2

(D) P $\longrightarrow$ 1;Q $\longrightarrow$ 2;R $\longrightarrow$ 4;S $\longrightarrow$ 3

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SECTION-II Integer : (Maximum Marks: 6)

Each question of this section have +3 marks.

1. Total number of yellow precipitates which is/are soluble in yellow ammonium sulphide (YAS).
CdS, Bi_2S_3, SnS, SnS_2, As_2S_3, As_2S_5, Sb_2S_3, Sb_2S_5, CuS, HgS

2. How many of the following cations gives ppt with H_2S in acidic solution :-
Zn^{2+} , Cu^{2+} , Ni^{2+} , Sr^{2+} , Al^{3+} , Bi^{3+} , Sn^{2+} , Pb^{2+}

SECTION-I (i) Single Correct (Maximum Marks:33)

Each question of this section have +3 marks

1. Given reactions take place during extraction of gold by hydrometallurgy.

$Au + CN^- + H_2O \xrightarrow{O_2} x$

$x + Zn \rightarrow y + Au$

x and y are respectively.

(A) $[Au(CN)_2]^-$ and $[Zn(CN)_4]^{4-}$

(B) $[Au(CN)_2]^{2-}$ and $[Zn(CN)_4]^{2-}$

(C) $[Au(CN)_4]^{3-}$ and $[Zn(CN)_6]^{4-}$

(D) $[Au(CN)_2]^-$ and $[Zn(CN)_4]^{2-}$

2. Copper is in metallic form in

(A) Schweizer's salt

(B) Bordeaux mixture

(C) German silver

(D) Chalcopyrites

3. Ore of a metal **M** (X) $\xrightarrow{\text{roasting}}$ Roasted ore $\xrightarrow[\text{absence of } O_2]{\Delta \text{ with (X)}}$ Metal **M**

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Ore of same metal **M** (Y) $\xrightarrow{\text{calcination}}$ Calcinated ore $\xrightarrow[\text{absence of } O_2]{\Delta \text{ with (X)}}$ Metal **M**

Select correct option :

(A) X = Chalcocite Y = Malachite

(B) X = Galena Y = Cerrusite

(C) X = Zinc blende Y = Calamine

(D) (A) and (B) both are correct

4. Which of the following method does not involve conversion of the anion in ore

(A) Roasting of zinc blende

(B) Magnetic separation of tin stone

(C) Leaching of argentite

(D) Calcination of malachite

5. Concentration of cassiterite ore is based on :-

(A) Adsorption phenomenon

(B) Froth floatation

(C) Electromagnetic seperation

(D) Liquification

6. In self reduction method for the extraction of metal from galena. Oxidising agent and reducing agent are respectively :

(A) SO_2 and S^{2-} (B) Dipositive cation of metal and S^{2-}

(C) Roasted ore and O^{2-} (D) ore and roasted ore

7. Calcination followed by carbon reduction is applicable for the extraction of metal from :

(A) Sphalerite (B) Dolomite

(C) Calamine (D) Galena

8. Smithsonite is American name of ..... ore (European name). Metal is extracted from Smithsonite by calcination followed by carbon reduction. European name of Smithsonite is

(A) Calamine (B) Bauxite

(C) Dolomite (D) Sphalerite

9. Which of the following metal oxide are not reduced commercially by carbon reduction method.

(A) Cr_2O_3 (B) Mn_3O_4

(C) ZnO (D) Both (A) and (B)

10. Zinc blende ore on roasting at above 850°C gives

(A) ZnS and $ZnSO_4$ (B) CO_2 and ZnO

(C) SO_2 and ZnO (D) Zn and SO_2

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11. Which of the following reaction is an example of reduction of calcined or roasted ore into metal

(A) Bauxite ore + $Na_2CO_3 \xrightarrow{\text{fused}} NaAlO_2$

(B) $Ag_2S + NaCN \rightleftharpoons Na[Ag(CN)_2] + Na_2S$

(C) $ZnO + C \rightarrow Zn + CO$

(D) $CuSO_4$ (aq.) + $\underset{\text{dust}}{Zn} \rightarrow ZnSO_4 + Cu$

SECTION-I (ii) : (Maximum Marks: 24)

Each question of this section have +3 marks

12. Select ore in which oxidation number of metal is +2 and produce carbondioxide on calcination -

(A) Galena (B) Fools gold

(C) Siderite (D) Malachite

13. Consider metallurgy of Ag from argentite and select correct answer.

(A) NaCN act as depressant

(B) Zn act as reducing agent

(C) NaOH act as leaching agent

(D) Zn produce $[Zn(CN)_4]^{2-}$ complex

14. Iron is not present in the form of sulphide in :

(A) Fool's gold

(B) Siderite

(C) Chalcopyrite

(D) Limonite

15. Which of the following reactions are example of thermite reduction.

(A) A thermite reaction may start if a ship (having Al parts) is a hit by a missile

(B) $Al_2O_3 + 3Mg \xrightarrow{\Delta} 3MgO + 2Al$

(C) $Fe_2O_3 + 2Al \xrightarrow{\Delta} 2Fe + Al_2O_3$

(D) $Cr_2O_3 + Al \xrightarrow{\Delta} 2Cr + Al_2O_3$

16.

Roasted silver ore + CN^- + H_2O

↓ O_2 (air)

'X' + OH^-

↓ Zn

'Y' + Ag

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Which of the following statement is correct for the above process.

(A) Co-ordination number of X is two

(B) 'X' and 'Y' both are diamagnetic

(C) Zn acts as a catalyst

(D) CN^- ion is a complex forming agent as well as acts as a reducing agent.

17. Select the correct matching :

(A) Pyro metallurgy : Extraction of Fe

(B) Electro metallurgy : Extraction of Al

(C) Hydro metallurgy : Extraction of Au

(D) none of these

SECTION-I (iii) Matching List : (Maximum Marks: 3)

Each question of this section have +3 marks

18. Match the folloiwng list :-

	List - I (Ore)		List - II (Reduction process involved in the extration)
(P)	SnO_2	(1)	Self reduction
(Q)	$CuFeS_2$	(2)	Carbon reduction
(R)	NaCl	(3)	Electrolytic reduction
(S)	MnO_2	(4)	Alumino thermite process

(A) P $\longrightarrow$ 1;Q $\longrightarrow$ 2;R $\longrightarrow$ 4;S $\longrightarrow$ 3

(B) P $\longrightarrow$ 2;Q $\longrightarrow$ 1;R $\longrightarrow$ 3;S $\longrightarrow$ 4

(C) P $\longrightarrow$ 2;Q $\longrightarrow$ 1;R $\longrightarrow$ 4;S $\longrightarrow$ 3

(D) P $\longrightarrow$ 1;Q $\longrightarrow$ 3;R $\longrightarrow$ 2;S $\longrightarrow$ 4

SECTION-II : (Maximum Marks: 3)

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Each question of this section have +3 marks.

1. How many of the following process of metallurgy is not associated with high temperature :
Froth floatation, Cupellation, Hydrometallurgical reduction, Self reduction

SECTION-IV Match the Column : (Maximum Marks: 8)

Each question of this section have +8 marks
(Partial marks of each correct is +2)

1. Match the following column :-

	Column - I (Metallurgical change)		Column - II (Properties)
(A)	$CaCO_3 \longrightarrow CaO + CO_2$	(P)	Non-redox change
(B)	$ZnS + O_2 \longrightarrow ZnO + SO_2$	(Q)	Endothermic process
(C)	$C + O_2 \longrightarrow CO_2$	(R)	Exothermic process
(D)	$Fe_2O_3.xH_2O \xrightarrow{\Delta} Fe_2O_3$	(S)	Redox change
		(T)	Calcination

RCT # 1

SECTION-I (i) Single Correct	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	C	D	A	A	C	B	D	C	C
	Q.	11									
	A.	B									
SECTION-I (ii) Multiple Correct	Q.	12	13	14	15	16	17	18			
	A.	A,B,C,D	B,C	A,C	A,C	A,D	A,B,C	C,D			
SECTION-III Integer	Q.	1	2	3							
	A.	4	4	5							
SECTION-IV Match the Column	Q.	1									
	A.	A->PQR,B->R,C->S,D->PRT									

RCT # 2

SECTION-I (i) Single Correct	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	C	B	D	B	C	C	B	A	B
	Q.	11									
	A.	A									
SECTION-I (ii) Multiple Correct	Q.	12	13	14							
	A.	A,D	C,D	A,B,C,D							
SECTION-I (iii) Paragraph (SOT)	Q.	15	16	17	18						
	A.	A	C	C	B						
SECTION-II Integer	Q.	1	2								
	A.	5.83	8.00								
SECTION-IV Match the Column	Q.	1									
	A.	A->PQT,B->PR,C->PQR,D->PRS									

Rankers Academy JEE

RCT # 3

SECTION-I (i) Single Correct	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	B	C	B	B	A	A	D	A	B
	Q.	11									
	A.	A									
SECTION-I (ii) Multiple Correct	Q.	12	13	14	15						
	A.	A,B,D	A,C,D	C,D	A,B,D						
SECTION-I (iii) Paragraph (MCQ)	Q.	16	17								
	A.	A,B,C	A,B,D								
SECTION-III Integer	Q.	1	2								
	A.	5	4								
SECTION-IV Match the Column	Q.	1									
	A.	A->PR,B->QRST,C->QRS,D->Q									

RCT # 4

SECTION-I (i) Single Correct	Q.	1	2	3	4	5	6	7
	A.	A	B	B	A	D	B	C
SECTION-I (ii) Multiple Correct	Q.	8	9	10				
	A.	A,B,C	A,B,D	A,C,D				
SECTION-I (iii) Paragraph (SOT)	Q.	11	12					
	A.	A	4					
SECTION-I (iv) Matching List	Q.	13						
	A.	C						
SECTION-II Integer	Q.	1	2	3	4	5		
	A.	4.00	3.00	5.50	5.91 or 5.92	-96.80		

RCT # 5

SECTION-I (i) Single Correct	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	A	B	D	C	C	D	D	D	C
	Q.	11	12	13	14						
	A.	A	D	A	B						
SECTION-I (ii) Multiple Correct	Q.	15	16	17	18	19					
	A.	A,B,C,D	C,D	A,C	A,B	A,B,C					
SECTION-II Integer	Q.	1	2								
	A.	1.00	4.00								

RCT # 6

Rankers Academy JEE

SECTION-I (i) Single Correct	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	D	B	A	A	B	C	B	B	A
	Q.	11	12	13	14	15					
	A.	D	C	C	A	D					
SECTION-I (ii) Multiple Correct	Q.	16	17	18							
	A.	B,C	A,B	A,B							
SECTION-I (iii) Paragraph (SOT)	Q.	19	20								
	A.	B	B								

RCT # 7

SECTION-I (i) Single Correct	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	D	D	D	C	B	B	C	B	A
	Q.	11	12	13	14	15	16	17	18	19	
	A.	D	C	C	C	B	C	D	B	D	
SECTION-I (ii) Multiple Correct	Q.	20									
	A.	A,B,D									

RCT # 8

SECTION-I (i) Single Correct	Q.	1	2	3	4	5	6	7	8
	A.	D	D	D	B	A	D	B	4
SECTION-I (ii) Multiple Correct	Q.	9	10	11	12	13	14	15	
	A.	A,B	A,B,C	A,B,C,D	C,D	A,B	A,C	A,B,C,D	
SECTION-I (iii) Paragraph (SOT)	Q.	16	17						
	A.	A	D						
SECTION-III Integer	Q.	1	2	3					
	A.	3	3	2					

RCT # 9

SECTION-I (i) Single Correct	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	B	C	C	C	C	C	A	B	B
SECTION-I (ii) Multiple Correct	Q.	11	12	13							
	A.	A,C,D	A,B,C,D	A,B,C							
SECTION-I (iii) Paragraph (SOT)	Q.	14	15	16	17						
	A.	D	A	B	C						
SECTION-I (iv) Matching List	Q.	18									
	A.	D									
SECTION-II Integer	Q.	1	2								
	A.	3.00	1.00								

RCT # 10

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SECTION-I (i) Single Correct	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	C	A	C	D	B	D	C	B	D
SECTION-I (ii) Multiple Correct	Q.	11	12	13	14	15					
	A.	A,B,C,D	B,D	B,C	A,C,D	A,B					
SECTION-I (iii) Matching List	Q.	16									
	A.	A									
SECTION-II Integer	Q.	1	2	3							
	A.	4.00	0.00	1.00							
SECTION-IV Match the Column	Q.	1	2								
	A.	A->T,B->PQRS,C->P,D->Q	A->Q,B->QS,C->Q,D->PRT								

RCT # 11

SECTION-I (i) Single Correct	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	C	B	D	B	D	B	C	D	C
	Q.	11									
	A.	C									
SECTION-I (ii) Multiple Correct	Q.	12									
	A.	A,D									
SECTION-I (iii) Paragraph (SOT)	Q.	13	14	15	16						
	A.	A	A	A	A						
SECTION-I (iv) Matching List	Q.	17									
	A.	C									
SECTION-II Integer	Q.	1	2	3							
	A.	3.00	6.00	4.00							

RCT # 12

SECTION-I (i) Single Correct	Q.	1	2	3	4	5	6	7	8
	A.	D	C	B	C	D	C	4	C
SECTION-I (ii) Multiple Correct	Q.	9	10	11					
	A.	A,C,D	B,D	A,C					
SECTION-I (iii) Paragraph (SOT)	Q.	12	13						
	A.	A	D						
SECTION-II Integer	Q.	1	2	3	4				
	A.	6.00	1.00	0.00	4.00				
SECTION-IV Match the Column	Q.	1	2						
	A.	A->PQR,B->S,C->PS,D->PT	A->QT,B->PT,C->S,D->RT						

RCT # 13

SECTION-I (i) Single Correct	Q.	1	2	3	4	5	6	7	8
	A.	A	C	B	C	D	A	D	C
SECTION-I (ii) Multiple Correct	Q.	9	10	11	12	13	14		
	A.	B,C	A,B,C,D	B,D	B,C,D	A,B,D	A,D		
SECTION-I (iii) Paragraph (SOT)	Q.	15	16	17	18				
	A.	B	C	D	C				
SECTION-I (iv) Matching List	Q.	19							
	A.	A							
SECTION-II Integer	Q.	1	2						
	A.	3.00	4.00						

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RCT # 14

SECTION-I (i) Single Correct	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	C	D	B	C	B	C	A	D	C
	Q.	11									
	A.	C									
SECTION-I (ii) Multiple Correct	Q.	12	13	14	15	16	17				
	A.	C,D	B,D	B,D	A,C,D	A,B	A,B,C				
SECTION-I (iii) Matching List	Q.	18									
	A.	B									
SECTION-II Integer	Q.	1									
	A.	2.00									
SECTION-IV Match the Column	Q.	1									
	A.	A->PQT,B->RS,C->RS,D->PQT									

Important Notes

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